



Massive Bio – Request for Information (RFI): RFI on Accelerating the Adoption and Use of AI in Clinical Care (RIN 0955-AA13)
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Public Comment on the 2026 Request for Information (RFI): RFI on Accelerating the Adoption and Use of AI in Clinical Care (RIN 0955-AA13)

Subject: Response to Request for Information (RFI): RFI on Accelerating the Adoption and Use of AI in Clinical Care

To the Office of The Director, National Cancer Institute:

Introduction

On behalf of [Massive Bio, Inc.](#), a leader in leveraging artificial intelligence to connect cancer patients to optimal treatments and clinical trials, thank you for the opportunity to provide input on the development of the 2026 **RIN 0955-AA13: RFI on Accelerating the Adoption and Use of AI in Clinical Care**. As Co-Founder and Chief Medical AI Officer, I witness daily the transformative potential of AI to revolutionize healthcare, particularly in the complex field of oncology. We strongly support the Administration's commitment – as outlined in Executive Order 14179 – to sustain and enhance America's AI leadership in order to promote human flourishing, economic competitiveness, and national security (*Removing Barriers to American Leadership in Artificial Intelligence – The White House*).

Our response is grounded in successful real-world implementation, demonstrating how AI can democratize access to advanced care and clinical trials. We fully endorse the Administration's vision for American AI leadership and propose a "Genesis Project" for health—a coordinated national effort to operationalize AI transformation in clinical care.

The Core Challenge

The primary barriers to AI adoption are no longer technological, but systemic: fragmented data silos, misaligned reimbursement incentives, regulatory uncertainty regarding liability, and the need for trust⁴. To unlock the potential of AI, HHS must align its "regulatory, reimbursement, and research & development levers" simultaneously.

Our Strategic Position

Massive Bio stands ready as a key implementation partner. Through our Reticulum Nexus platform and partnerships with the American Cancer Society (ACS), we have proven that AI can be deployed responsibly at a national scale to break down silos between academic centers and community care. We urge HHS to move from strategy to execution, launching demonstration projects that propel the U.S. healthcare system forward.

Key Recommendations for HHS

Accelerating AI in Clinical Care: A "Three-Lever" Approach

To transition AI from "bench to bedside," Massive Bio recommends HHS prioritize the following high-impact changes:

1. Modernize Regulation & Governance (The Safety Lever)

- **Update FDA Frameworks:** Finalize the "Total Product Lifecycle" approach for Software as a Medical Device (SaMD), allowing AI algorithms to learn and improve over time without requiring new submissions for every update.
- **Clarify Liability via Safe Harbors:** Establish clear guidance that clinicians remain the final authority, but provide legal "safe harbors" for providers who follow recommendations from HHS-certified or validated AI tools.
- **Voluntary AI Certification:** Launch an ONC-administered voluntary certification program for clinical AI tools to validate safety, bias mitigation, and interoperability, giving providers the confidence to adopt them.

2. Align Reimbursement & Incentives (The Motivation Lever)

- **Create CPT Codes for AI Services:** Introduce billing codes for "AI-assisted decision support" and care coordination to reimburse the time and complexity of using AI tools that improve outcomes.



- **Reward Outcomes in Value-Based Care:** Explicitly incorporate AI usage into Alternative Payment Models (APMs) and MIPS. For example, award bonus points to ACOs that use AI to identify high-risk patients or match cancer patients to clinical trials¹³.
- **Modernize Prior Authorization:** Encourage a "gold card" system where AI-driven requests that meet evidence-based criteria receive automatic, real-time approval, reducing administrative burden¹⁴.

3. Strengthen Infrastructure & Research (The Enabler Lever)

- **Enforce Interoperability Standards:** Double down on **FHIR** and **mCODE** adoption to break down data silos. Mandate "bulk data" access capabilities to fuel AI research and ensure diverse, representative training datasets.
- **Prioritize Implementation Science:** Shift research funding toward "implementation science"—studying *how* to integrate AI into clinical workflows effectively—rather than just algorithm development¹⁶.
- **Establish a National AI Testing Sandbox:** Create a public-private "validation lab" where vendors can test algorithms on federated, de-identified datasets to benchmark performance and fairness without compromising patient privacy.

Conclusion

By pulling these levers, HHS can build an ecosystem of trust and utility. Massive Bio is prepared to collaborate with HHS, offering our technology and operational expertise to turn this vision into a reality for American patients.

Introduction

Massive Bio appreciates the opportunity to comment on the Department of Health and Human Services (HHS) Request for Information, *"Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care"* (RIN 0955-AA13). As an AI-driven oncology company, Massive Bio is uniquely positioned to offer bold, visionary insights aligned with President Trump's agenda for American AI leadership and HHS's transformative goals^{[1][2]}. Our perspective is grounded in real-world implementation: Massive Bio has pioneered patient-consented, patient-centric oncology data aggregation and AI analytics at national scale. We have demonstrated how artificial intelligence can **propel the U.S. health care system forward** by improving patient outcomes, reducing costs, and expanding access to cutting-edge care^{[3][2]}. We stand ready as a key implementation partner to HHS, particularly in cancer care, to operationalize the Administration's bold vision of AI-powered healthcare delivery.

About Massive Bio: Founded in 2015, Massive Bio has become a leader in AI-enabled cancer care, serving over 160,000 patients to date across all 50 states and globally^{[4][5]}. We collaborate with the American Cancer Society (ACS) and other stakeholders to break down silos between academic centers and community care, bringing **AI-driven solutions directly to patients and providers wherever they are**. For example, Massive Bio's technology powers **ACS ACTS** – a nationwide program helping patients of any background find personalized clinical trial options – providing AI-matched trial lists and support resources to **patients in every state**^{[6][7]}. Our **Reticulum Nexus** platform, unveiled at the ESMO 2025 Congress, is an AI-driven, multi-agent system that **orchestrates the end-to-end oncology journey** from initial patient outreach and consent to data analytics, trial matching, and last-mile enrollment logistics^{[8][9]}. Reticulum Nexus connects previously fragmented data sources through native **FHIR/mCODE interoperability**, linking electronic health records (EHRs), labs, sponsors, community clinics and registries on one fabric^[10]. What once took hundreds of hours of manual effort now runs as an **auditable, human-in-the-loop AI workflow**, resulting in *"speed with empathy: more patients reached, faster answers, and auditable decisions"*^[9]. We have proven that AI can be deployed responsibly at national scale **outside of academic silos**, as evidenced by our partnership with ACS and recognition by the White House Cancer Moonshot for our AI-powered trial pre-screening hubs^{[11][12]}. These efforts have **dramatically shortened trial enrollment timelines and boosted participation rates** by proactively identifying eligible patients through genomic and clinical data^[12]. Importantly, Massive Bio's approach keeps patients at the center: data is used only with patient consent, de-identified for research insights, and integrated with *"privacy-by-design"* governance to maintain trust^{[10][13]}. Our mission is to **democratize access to advanced treatments** – using AI to open clinical trials to diverse populations across geography, race, age, and socioeconomic status – while accelerating **value-based care** through precision targeting of therapies to those who will benefit most^{[14][15]}. We leverage real-world data (RWD) from thousands of patients to generate evidence and optimize trial design, all in alignment with HHS's "OneHHS" strategy to extend AI's benefits across the health system^{[16][17]}.

In this comment, we respond to each of the ten questions posed in the RFI. For each, we provide concrete answers grounded in Massive Bio's experience and supported by real examples. We also offer forward-looking recommendations on how HHS can use its **regulatory, reimbursement, and research & development levers**^{[3][17]} to unlock AI's impact in oncology and beyond. Central themes of our responses include: modernizing regulation to foster innovation while protecting patients; aligning reimbursement and value-based payment models to reward AI-driven improvements; investing in interoperability and data



standards to fuel AI development; and ensuring that evidence generation and **implementation science** keep pace with technology. We frame our suggestions in the spirit of HHS's **Genesis Project** – analogous to a health-focused “moonshot” – to bring **operational AI transformation to clinical reality** on a national scale. Massive Bio stands ready to partner with HHS to deliver on this ambitious agenda, translating policy into on-the-ground results for American patients.

Question 1. Biggest Barriers to Private-Sector AI Innovation and Adoption in Clinical Care

Q1: “What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?”

Key Barriers: Despite rapid advances in AI, private-sector innovators face significant hurdles bringing AI tools from bench to bedside. Major barriers include:

- **Fragmented Data and Interoperability Challenges:** Quality healthcare AI depends on large volumes of diverse data, but health data today is siloed across EHR systems, labs, imaging archives, and devices. Data fragmentation – and the lack of seamless interoperability – is a top barrier to AI innovation. Many AI developers struggle to obtain diverse, representative datasets due to information blocking or inconsistent standards. For example, oncology data lives in disparate systems (EMRs, genomic testing labs, clinical trial databases) that historically do not communicate. This fragmentation hinders training robust AI models and limits an AI tool's ability to *ingest real-time patient data* during clinical use. Massive Bio has confronted this challenge directly: our AI platform integrates data from community clinics, academic centers, and patient-reported sources using common standards (FHIR, mCODE)^[10]. However, not all developers have the resources or permissions to aggregate such data. HHS's focus on “**data liquidity**” and **patient data trust** is well placed – improving interoperability is essential to overcome this foundational barrier^[18].
- **Regulatory Uncertainty and Risk Aversion:** Innovators face ambiguity about how existing health regulations apply to AI, especially software not clearly classified as medical devices. Uncertainty around FDA oversight, HIPAA, and liability for AI-driven decisions creates a chilling effect. Startups fear investing in a tool that might later be deemed a regulated device requiring lengthy approval, while providers fear using AI without clear legal guidelines. For instance, if an AI clinical decision support system suggests a treatment and an error occurs, it's unclear who is liable – the clinician, the hospital, or the software maker?^[19] This uncertainty leads to **risk-averse institutional cultures**, where hospital compliance teams often err on the side of caution and delay adoption of new AI tools. Clear guidance is needed to assure innovators and users that they can deploy AI safely within existing legal frameworks (more on this in Q3). Until then, many promising AI solutions remain stuck in pilot phases. HHS's RFI aptly notes that *digital health regulatory frameworks must evolve to account for AI-driven tools while maintaining safety*^[17] – the private sector needs this evolution to proceed confidently with innovation.
- **Lack of Reimbursement and Misaligned Incentives:** The current payment landscape does not reward – and can even penalize – the use of AI in clinical practice. Providers have little financial incentive to adopt AI tools that, for example, prevent hospitalizations or automate tasks, because fee-for-service reimbursement is tied to volume of services. An AI that improves efficiency or avoids an expensive therapy might reduce revenue under traditional models. Meanwhile, there are few billing codes or payment models that directly reimburse AI-driven care improvements (e.g. no codes for “AI-assisted diagnosis” or “AI care coordination”). The absence of reimbursement pathways is repeatedly cited by providers as a barrier to investing in AI^[20]. This is particularly relevant in oncology: many AI tools could help identify the most cost-effective therapy or an appropriate clinical trial (saving costs of ineffective treatments), but providers face upfront costs to integrate these tools without immediate reimbursement. Additionally, **prior authorization processes** can unintentionally discourage innovation – if an AI recommends a novel therapy, insurers may not approve it without extensive documentation, delaying care. Massive Bio's approach of offering clinical trial options *before* prior authorization barriers arise is one way to sidestep this, ensuring patients can access alternative cutting-edge treatments without waiting for insurance approvals. However, broader payment reform is needed so that AI that **reduces costs and improves outcomes is financially rewarded**, not discouraged. We discuss specific reimbursement levers in Q2.
- **Clinical Integration and Workflow Challenges:** Integrating AI into clinical workflows remains difficult. Busy healthcare providers are often overwhelmed with technology (EHR alerts, documentation burdens), and a new AI tool can feel like added complexity unless it is seamlessly embedded. If AI outputs are not presented at the right time and in an intuitive way, clinicians may ignore them. Training requirements can also be a barrier – clinicians need to trust and understand the AI, which takes education. In smaller practices or rural hospitals, there may be limited IT support to implement advanced AI systems^[21]. This creates an **adoption gap**: large health systems with resources charge ahead, while smaller providers lag behind, exacerbating disparities^[21]. Massive Bio has observed this gap in oncology; community oncologists often lack the decision support tools available at major cancer centers. Our solution has been to provide AI-driven trial matching as a *turn-key service* accessible via a simple web interface or phone call, which lowers the barrier for small practices to participate. Nonetheless, widespread clinical adoption will require HHS to support easier integration



(through standards and technical assistance) and to cultivate an AI-ready workforce. Change management and training are as important as the technology itself – clinicians need to see AI as a helpful colleague, not a hindrance.

- **Trust, Transparency, and Bias Concerns:** Finally, a pervasive barrier is the concern among both providers and patients about the trustworthiness of AI. Healthcare is fundamentally human; if AI is seen as a “black box” making life-and-death suggestions, clinicians may be reluctant to rely on it. Past high-profile failures have made some skeptical – for example, early systems like IBM’s Watson for Oncology did not live up to the hype, undermining confidence in AI’s capabilities. Additionally, well-publicized studies have shown that some healthcare algorithms had **biases that could perpetuate disparities**, for instance under-representing Black patients in care management programs due to biased training data[22][23]. These issues create caution in adoption: providers worry about safety and bias, and patients worry about fairness and privacy. For the private sector, this means extra effort is needed to demonstrate algorithm validation, fairness, and explainability. Massive Bio has invested heavily in transparency – our trial matching AI provides **clinician-readable rationales** for why a patient is eligible for a trial, and our “Radiant Core” control plane ensures every AI decision is auditable for quality and compliance[24][25]. We have also focused on diversity in our data (partnering with organizations like the **American Cancer Society** and community clinics) so that our AI recommendations are equitable and inclusive[15][26]. Building trust through such measures is crucial. HHS can assist by establishing or endorsing standards for AI transparency and bias mitigation, giving providers and the public greater confidence to embrace AI tools.

In summary, the biggest barriers to private-sector AI innovation and adoption in health care are not technological – they are systemic. **Data silos, unclear regulations, misaligned payments, workflow integration challenges, and trust issues** all conspire to impede progress. These barriers are interrelated and require a coordinated strategy across HHS agencies (ONC, CMS, FDA, OCR, etc.) to address. We are encouraged that HHS recognizes many of these challenges in this RFI. By systematically lowering these barriers – as outlined in the following responses – HHS can unleash a wave of private-sector innovation, allowing AI to fulfill its promise of improving clinical care.

Question 2. Regulatory, Payment Policy, and Programmatic Changes to Incentivize AI in Clinical Care

Q2: “What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.”

Massive Bio believes HHS can accelerate AI adoption by acting decisively on three fronts – **regulation, reimbursement, and program design** – the very levers highlighted in the RFI[3][17]. We recommend the following priority changes:

A. Evolve Regulatory Frameworks to Foster Innovation with Safety:

1. **Modernize FDA’s Digital Health Oversight:** HHS should work with the Food and Drug Administration (FDA) to update guidance on AI/ML-based software in clinical care. Current FDA regulations (21 CFR Part 820 and related guidances) were not designed with continuously learning algorithms in mind. We urge HHS to finalize and implement the FDA’s proposed framework for AI/ML SaMD (Software as a Medical Device) that allows for a “**total product lifecycle**” approach. This would let developers make certain algorithm updates under pre-agreed protocols (e.g. a “Predetermined Change Control Plan”) without requiring new submissions each time. Such an approach maintains safety and effectiveness while allowing AI tools to improve over time. Additionally, clarification is needed on what constitutes a regulated “medical device” versus clinical decision support exempt from regulation under 21st Century Cures Act Section 3060. HHS should issue guidance (or support FDA in doing so) that clearly delineates the boundary – for example, predictive analytics that drive therapy decisions might be regulated, whereas administrative AI or transparent decision support might not. Clear rules will **reduce uncertainty and spur innovation**, as companies will know whether to pursue FDA clearance or not. We also suggest HHS consider **regulatory sandboxes or pilot programs** in partnership with FDA where novel AI solutions (especially those addressing high-need areas like cancer) can be tested in a controlled setting with temporary regulatory flexibility. This can generate evidence to inform permanent regulatory changes.
1. **Address Liability and Safe Harbors:** To encourage providers to adopt AI, HHS should explore creating safe harbors or guidance around liability when using approved AI tools. Fear of malpractice exposure is a deterrent to clinical AI use. For example, if a physician follows an AI recommendation that is consistent with clinical guidelines or an HHS-endorsed tool and an adverse event occurs, that physician should not be held to a higher liability standard than if they had followed standard practice. HHS, possibly via the Agency for Healthcare Research and Quality (AHRQ) or in collaboration with professional societies, could develop “*best practice*” *guidelines* for AI-assisted care. Providers adhering to those could be granted certain protections (similar in concept to safe harbors for following evidence-based guidelines). Clarifying liability can reduce the “**novel legal issues**” noted in the RFI[19]. Likewise, HHS could encourage states to update medical malpractice laws or insurer policies to recognize the role of AI. Additionally, we recommend HHS consider working with

the Office of Inspector General (OIG) to ensure that **Stark Law and Anti-Kickback** regulations do not inadvertently impede AI tool adoption. For instance, hospitals might want to provide an AI decision support tool to community physicians to improve care coordination, but worry this could be seen as remuneration. Providing exceptions or guidance for sharing AI tools that improve quality (akin to the EHR donation safe harbor) would help spread AI to smaller practices.

2. **Strengthen Data Privacy & Sharing Policies for AI:** Privacy regulations should be updated to facilitate responsible AI development while protecting patients. HHS's Office for Civil Rights (OCR) could clarify how HIPAA permits the use of de-identified data for algorithm training and explicitly allow patients to authorize use of their data for AI R&D. Many patients are willing to contribute data to improve care as long as it's secure and anonymous – policies should empower that. We recommend HHS develop standard **patient consent frameworks** for AI (potentially building on the model of research biobanks) so that private companies can create *opt-in patient data repositories* for AI training under HHS oversight. Furthermore, HHS should promote the adoption of *Privacy Enhancing Technologies* (like federated learning and differential privacy) by issuing guidance or even amending the HIPAA Privacy Rule to recognize these techniques as safeguards. For example, allowing health systems to participate in federated model training (where data doesn't leave their servers) without it being considered a HIPAA disclosure could accelerate cross-organization data collaborations. In summary, privacy rules should not be static barriers but rather enablers of **"safe, secure, and allowed" data use for AI**^[18]. By updating regulations to explicitly accommodate modern AI data practices, HHS can both protect individuals and advance innovation.

B. Align Reimbursement and Payment Policy with AI-Driven Care:

1. **Create Reimbursement Pathways for AI Services:** CMS should establish clear reimbursement mechanisms for clinical uses of AI that add value. We recommend developing new CPT codes or modifiers for AI-assisted procedures and decision support. For example, if an oncologist uses an AI platform to identify a therapy or clinical trial for a patient, there could be a billable code for the time and complexity involved in that AI-informed care coordination. Similarly, radiologists who utilize AI to read scans or pathologists using AI for slide analysis should have that reflected in billing (e.g., an add-on code indicating "AI augmentation used"). These codes could initially be introduced as Category III (tracking) codes to collect data, then progress to payment once value is demonstrated. Additionally, CMS's technology add-on payments (NTAP) for new innovations might be extended to AI: if a hospital adopts an AI system that demonstrably reduces complications or readmissions, an add-on payment could offset the cost. The underlying principle is to **reward outcomes and efficiency** – if AI reduces total cost of care or improves quality, providers should share in the savings. In oncology, for instance, if an AI tool helps avoid ineffective chemotherapy by matching a patient to a targeted therapy trial at no cost, the payer saves money and the patient potentially has better outcomes. CMS could capture this by allowing some form of shared savings or payment for the care coordination that led to trial enrollment. Without such reimbursement, providers may hesitate to take the extra steps to use AI for trial matching, even though it benefits the system overall. We applaud HHS for explicitly asking how reimbursement structures can be *"simplified and better aligned to support...efficient, deflationary technologies"*^[17]. One specific policy to revisit is Medicare's reimbursement for telehealth and remote monitoring: as AI becomes embedded in these services (e.g., AI chatbots for patient follow-up, AI-enhanced remote patient monitoring for symptoms), CMS should ensure coverage and payment parity. The recent expansion of telehealth codes could be expanded further to cover **"AI-assisted remote care"** so long as there is oversight by a clinician. Likewise, CMS might consider demo programs where it reimburses AI-driven preventive care interventions that reduce hospitalizations (for example, paying a monthly fee for an AI predictive service that helps manage high-risk patients, similar to chronic care management fees).

1. **Incorporate AI into Value-Based Care Models and Quality Programs:** HHS should leverage alternative payment models (APMs) to drive AI adoption. Accountable Care Organizations (ACOs), bundled payment programs, and specialty models like the Oncology Care Model (OCM) are natural vehicles for AI since they reward outcomes and cost reduction. We recommend that CMS explicitly encourage or even require the use of advanced analytics/AI in these models. For example, in a future Oncology Value-Based model, CMS could include quality measures or bonus points for utilizing evidence-based AI decision support that improves care (like matching stage IV cancer patients to trials or genomic-guided therapies). CMS Innovation Center (CMMI) could pilot an "AI Accelerator" within ACOs – perhaps giving participating ACOs **technical assistance grants to implement AI tools** for care coordination, and then measuring the impact on cost and quality. If successful, that could be scaled. Furthermore, HHS can align Medicare Advantage and Medicaid policies to support AI: for instance, allowing plans to count investments in AI tools as quality improvement activities in their Medical Loss Ratio (MLR) calculations, thereby incentivizing insurers to invest in provider-facing AI. On the quality measurement front, HHS could work with organizations like NCQA and The Joint Commission to incorporate AI-enabled care processes into accreditation and scoring. A concrete near-term step: include use of clinical AI in Merit-Based Incentive Payment System (MIPS) Improvement Activities. Clinicians could earn credit for implementing an AI tool that, say, identifies care gaps or reduces medication errors. Aligning quality incentives will nudge providers toward AI as a means to achieve targets.

2. **Reduce Administrative Barriers and Prior Auth Burdens via AI:** We suggest HHS revise policies that inadvertently discourage AI by imposing unnecessary administrative hurdles. A prime example is prior authorization. HHS can modernize prior auth regulations to recognize AI-based determinations. For instance, if an AI algorithm (certified for accuracy) finds that a patient meets evidence-based criteria for a certain test or referral, payers could be required to **automatically approve** it or use a “gold card” approach for providers using such tools. This would flip prior auth into a more automated, AI-driven process that reduces delays. CMS could pilot this in Medicare Advantage: encourage MA plans to integrate AI for initial screening of auth requests, granting immediate approval when criteria are met (perhaps tying this to the recently proposed rules on electronic prior auth to ensure timely decisions). By doing so, **patients get needed care faster** and the burden on providers is lessened^[27]. From Massive Bio’s perspective, offering patients clinical trial alternatives is one way to avoid the prior auth maze altogether – if HHS were to formally support policies that “*clinical trial enrollment can be considered a prior auth exception*” (for instance, pausing prior auth on standard treatment when a patient opts for an eligible clinical trial), it would be a strong incentive for providers to utilize AI matching services proactively. Essentially, reward pathways that solve the prior auth challenge either by automation or by providing alternative options (like trials) that render prior auth moot.

C. Programmatic and Policy Design Changes:

1. **Expand Public-Private Partnerships and Challenge Programs:** HHS should use its convening power to create partnerships that accelerate AI implementation. One idea is an **AI Clinical Care Challenge** series – prize competitions (as authorized by the America COMPETES Act) around specific needs (e.g., “AI for Cancer Detection Challenge” or “AI for Reducing Hospital Readmissions”). These could be run by HHS’s IDEA Lab or ONC, with winners receiving funding or pilot opportunities at HHS hospitals (like the VA or IHS)^[28]. Such competitions signal HHS’s priorities and incentivize private innovators to focus on high-impact areas. Additionally, HHS could establish *Centers of Excellence for AI in Clinical Practice* via grant funding (perhaps through AHRQ or HRSA for community health centers). These centers (academic or community consortiums) would develop, test, and disseminate best practices for AI adoption – effectively demonstration sites that others can learn from. For example, Massive Bio is a founding member of the federally sponsored **CancerX** public-private partnership^{[29][30]}; continued support and funding for such consortiums will drive multi-stakeholder solutions to common barriers. We recommend HHS also collaborate with the Department of Energy and NSF on the **Genesis Mission** initiative – while DOE’s Genesis Mission focuses on AI for scientific research^{[31][32]}, a parallel HHS “Genesis Project” could focus on AI for health challenges, ensuring healthcare is a key domain in national AI strategy. By aligning with broader national AI missions, HHS can tap into additional resources and talent for healthcare AI.

1. **Revise ONC Certification and CMS Conditions of Participation:** ONC’s EHR certification program has been effective in standardizing certain functionalities. HHS could consider an ONC-administered **AI Certification program** – voluntary at first – where AI clinical tools are assessed for basic safety, interoperability, and bias mitigation. If a tool meets the criteria, it gets an HHS “stamp of approval,” giving providers confidence to adopt it. This kind of certification could later tie to CMS conditions (e.g., hospitals using AI for clinical decisions must use certified tools to qualify for Medicare reimbursement, ensuring quality). Similarly, CMS could update hospital Conditions of Participation to require an **AI governance plan** for hospitals deploying AI (ensuring they address bias, monitor outcomes, etc.). While adding requirements may seem like a burden, done thoughtfully it can actually encourage hospitals to embrace AI with proper oversight rather than avoid it.
2. **Revisit CFR provisions hindering data sharing:** HHS might identify specific regulations in 42 CFR (Medicare/Medicaid) or 45 CFR (Public Welfare, including HIPAA rules) that unnecessarily restrict innovative uses of AI. For example, 42 CFR Part 2 (confidentiality of substance use records) is one area that, while important, complicates integrating behavioral health data into AI models; aligning Part 2 with HIPAA as recently proposed will help comprehensive AI care. Another example: 45 CFR 164 (HIPAA Privacy Rule) – defining de-identification standards that allow for the sharing of key data like genomic information (which currently can be tricky under the “identifier” removal rules). Clarifying that codified genomic variants or certain derived data can be shared if individuals cannot be readily identified would help AI in precision medicine. We encourage HHS to take a holistic look at its regulations to find and fix such friction points.

In summary, HHS should **act boldly to update regulations, realign incentives, and design programs** that collectively turbocharge AI adoption in care. By prioritizing the changes above, HHS will send a strong signal that America is committed to operationalizing AI for health. These steps will incentivize companies like Massive Bio to continue innovating in areas like oncology, knowing that a supportive policy environment is in place. Importantly, none of these changes sacrifice patient safety or privacy – rather, they **modernize protections and payments to fit AI**, ensuring that the effective use of AI is rewarded and scaled. Massive Bio is especially enthusiastic about these changes because they would directly augment our ability to develop and deploy AI for clinical trials and precision oncology. For example, clearer FDA guidance would streamline enhancement of our trial-matching algorithms; new reimbursement for AI-driven care coordination would allow more clinics to adopt our platform; and partnership programs would enable us to demonstrate our tech in more public settings. We believe these policy moves are not only justified – they are urgently needed to keep pace with technology and to realize AI’s potential to “*propel the American health care system forward.*”^[3]

Question 3. Novel Legal and Implementation Issues for Non-Medical-Device AI – and HHS’s Role

Q3: “For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?”

AI tools that are **not regulated as medical devices** (e.g., clinical support software, administrative AI systems, etc.) indeed present new challenges for our health care legal and governance frameworks. Massive Bio engages with several such AI applications (for example, our AI-driven patient triage and scheduling tools are not FDA-regulated devices but are used in care workflows), so we have first-hand insight into these issues. Key novel issues and recommended HHS roles include:

- **Liability and Accountability for AI-Influenced Decisions:** When an AI system influences clinical decisions, it blurs traditional lines of accountability. If an algorithm (say, an AI risk score or triage recommendation) contributes to a clinical decision that leads to harm, there is ambiguity over who is legally responsible – the clinician who accepted the AI’s advice? The hospital that deployed the tool? The software developer? This lack of clarity is a “novel” issue because our malpractice system assumes human decision-makers. HHS should convene stakeholders (providers, developers, insurers, patient advocates) to develop consensus guidelines on AI liability. While legislation may ultimately be needed, in the near term HHS can issue guidance that clarifies expectations. For instance, HHS could endorse the approach that **clinicians remain the final authority** and thus primarily liable, but that following validated AI recommendations could be considered in determining the standard of care. Simultaneously, HHS could work with professional liability insurers to encourage coverage models that don’t penalize providers for using AI that improves care. By framing when it is appropriate to rely on AI (e.g., if tool meets certain validation criteria), HHS can give providers confidence to use AI without fear of unjust legal exposure. Additionally, HHS might explore model legislation for states that addresses AI liability – for example, laws that provide a rebuttable presumption that if a clinician followed an HHS-certified AI recommendation, they met the standard of care. This would address liability concerns in a balanced way.
- **Indemnification and Vendor Responsibility:** Hospitals and health systems often seek indemnification from AI vendors for any harm caused by their software. However, vendors are wary of open-ended liability, especially since the end-users ultimately make clinical decisions. This dynamic can slow adoption – negotiations over indemnity can be lengthy or some vendors refuse to enter the healthcare market. HHS could help by developing **model contract language or best practices for AI procurement** in healthcare, perhaps through ONC or AHRQ. By standardizing how responsibility is shared (e.g., vendor liable for software faults, provider liable for misuse), HHS can ease the burden on individual organizations to sort this out. Another approach is HHS fostering the creation of an **insurance pool or fund for AI-related incidents** – analogous to the National Vaccine Injury Compensation Program, which removed direct manufacturer liability as a barrier to vaccine availability. A similar concept could be considered for certain AI: an industry-wide fund that compensates patients in rare cases of AI error causing harm, thereby limiting the need for each vendor to carry massive liability insurance. HHS’s role would be in studying feasibility and potentially advocating for enabling legislation if needed.
- **Privacy and Data Use Ethics:** Unlike traditional medical devices, many AI systems feed on vast datasets – including patient information – raising concerns about privacy beyond HIPAA’s original scope. Two novel concerns are: (1) AI models trained on patient data might inadvertently “remember” sensitive details (creating risk of re-identification or unintended disclosure), and (2) companies may seek to use patient data for secondary AI development in ways patients don’t expect. To address these, HHS should update privacy regulations and guidance as discussed in Q2. Specifically, HHS/OCR should clarify how de-identification rules apply to complex data like genomic or free-text data used in AI. It may be necessary to develop **new guidelines or safe harbors for AI model training**, ensuring that if certain best practices are followed (like data aggregation, removal of direct identifiers, bias testing to protect minority groups), the use is permissible. HHS might also support development of **technical standards for AI model security** – for instance, encouraging the use of encryption, access controls, and audit trails for AI systems that handle patient data. The idea of AI systems themselves needing to be HIPAA-compliant is novel. Perhaps HHS could assert that significant AI models should undergo a *Privacy Impact Assessment* or be certified for how they handle data. In essence, HHS should extend its existing privacy and security frameworks to explicitly cover the AI lifecycle (data collection → model training → deployment → data outputs). This might include guidance on issues like using cloud-based AI services (which introduces third-party concerns) or leveraging patient data to create synthetic data for AI (potentially a privacy-preserving practice HHS could endorse).
- **Bias, Fairness, and Health Equity Concerns:** Not specifically legal in the traditional sense, but a governance issue: AI tools can inadvertently propagate bias against protected classes (race, gender, etc.), leading to disparities in care. While anti-discrimination laws (like Civil Rights Act Title VI for entities receiving federal funds) could be interpreted to cover biased AI, this is uncharted territory. HHS should take an active role in setting expectations that AI in clinical settings must be

fair and unbiased. We suggest HHS, through the Office of Civil Rights and Office of Minority Health, issue guidance or even regulations requiring that any AI used in clinical care undergo **bias testing and mitigation**. This could involve routine audits to see if outcomes differ by race or other factors. If significant bias is found and not corrected, that could be considered non-compliance with, say, healthcare civil rights obligations. HHS might incorporate this into Medicare Conditions of Participation or certification criteria. The role of HHS here is to ensure accountability: just as hospitals must have infection control programs, in the future they should have **algorithmic bias control programs**. This is a new frontier, but one where HHS leadership is crucial to prevent AI from widening disparities. Massive Bio, for instance, proactively designs its trial matching AI to expand options for underserved populations^[15]. We welcome HHS's help in codifying such practices industry-wide.

- **Transparency and Explainability Requirements:** A novel implementation issue is how to govern “black box” algorithms. Traditional medical practice relies on explainable logic (doctors can be asked why they chose a treatment). With AI, if neither the user nor the patient can understand the basis for a recommendation, it challenges informed consent and accountability. HHS should consider policies that push for **algorithmic transparency** appropriate to context. For non-device AI, which isn't going through FDA premarket review, this could mean requiring that patients be informed when AI is involved in their care and, upon request, given a basic explanation of how it works (similar to how patients have a right to explanations of their care plan). Moreover, HHS can promote standards for explainability – e.g., if an AI recommends a cancer trial, it should provide the oncologist with the key reasons (patient has X mutation, Y clinical criteria matched) in human-readable form. This is something Massive Bio has implemented (transparent rationales)^[33], and it greatly aids clinician acceptance. HHS might not need a new law for this; even a guidance or adding a requirement in Medicare Advantage plan oversight (since MA plans are increasingly using AI for utilization management) that beneficiaries are informed of AI use would set a precedent.

In all the above areas, **HHS's role is to provide guidance, standards, and a forum for resolution** of these novel issues. The Department can use its influence to shape soft law (guidelines, best practices) that preempt the need for constantly reacting in courts after problems occur. We specifically recommend HHS establish a **Healthcare AI Governance Council** – bringing together FDA, ONC, CMS, OCR, OIG, and external experts – to continuously monitor these emerging issues and update policies. This council could issue annual reports or recommendations to keep regulations nimble. The council could tackle questions like: Should there be an “AI Incident Reporting System” (analogous to FDA's adverse event reporting, but for AI errors or near-misses)? How do we harmonize state laws (e.g., some states may start passing their own rules on AI liability or data use – HHS could offer a model to encourage consistency)?

Massive Bio's perspective is that sensible guardrails will ultimately **increase adoption** of AI, by ensuring trust and accountability. HHS should neither micromanage nor take a hands-off approach, but rather serve as an **enabler and coordinator** on these governance challenges. By addressing liability, privacy, bias, and transparency in a proactive way, HHS will help remove these as points of friction or fear. For example, once providers know HHS's stance on AI-related liability and see a clear framework, they will be more willing to integrate AI (as one physician told us, “I just want to know I won't be hung out to dry if I use the AI recommendation in good faith”). Similarly, by ensuring AI adheres to privacy and equity standards, HHS will build public trust, which is foundational for widespread AI use. In summary, HHS should play an active role by updating governance structures to accommodate AI, thereby protecting patients and providers while unleashing the technology's benefits.

Question 4. Promising Evaluation Methods for Non-Device AI – and How HHS Can Support Them

Q4: “For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?”

Evaluating AI tools that are not subject to FDA pre-market review is a critical challenge. Without a formal regulatory approval process, we need other methods to ensure these AI systems are safe, effective, and fit seamlessly into clinical workflows. Fortunately, the industry and research community have been developing evaluation approaches. Massive Bio has practical experience in this area – we continuously evaluate our AI matching algorithms both **before deployment (in silico and pilot testing)** and **after deployment (real-world performance monitoring)** to ensure they function as intended for patients and clinicians. Below, we outline promising evaluation methods and suggest how HHS can bolster them:

Promising Pre-Deployment Evaluation Methods:

- **Retrospective Validation with Gold-Standard Datasets:** Before an AI is used in live clinical care, one robust method is to test it on retrospective data where outcomes or “ground truth” decisions are known. For example, an AI that recommends cancer clinical trials can be validated on past patient cases: feed in historical patient data and see if the AI would have identified the trials that those patients actually enrolled in (or that experts would have chosen). Similarly,

diagnostic AIs can be tested against existing datasets of images or cases with known diagnoses. These validations measure accuracy, sensitivity, specificity, etc. A promising trend is the creation of **open clinical datasets and challenges** – for instance, the NIH has released large imaging datasets for AI algorithm benchmarking. Community competitions (such as MICCAI challenges in imaging) have driven algorithm improvements. HHS can encourage this by supporting the curation and sharing of de-identified clinical datasets as evaluation benchmarks. Metrics like area-under-curve (AUC) for accuracy, or positive predictive value, are common, but evaluation should go beyond just accuracy: robustness across subgroups (e.g., how does performance vary by age, ethnicity, hospital site?) is key. Pre-deployment robustness testing might involve intentionally **stress-testing** the AI with edge cases or noisy data to see if it still performs. For example, at Massive Bio, we test our trial matching AI on data with missing fields or atypical patient profiles to ensure the system can handle real-world messiness.

- **Simulation and Clinical Vignettes:** Another pre-deployment approach is simulation studies. In a controlled setting, clinicians can be presented with cases (vignettes) and the AI's recommendation, and researchers measure how the clinicians respond, whether the AI's input improved decision quality or speed. Think of it as a "clinical Turing test" – does the clinician aided by AI do better than without? This human-centered evaluation assesses usability and usefulness, not just technical accuracy. For instance, we have conducted user-acceptance testing with oncologists for our AI interface: we simulate a patient consult with and without the AI suggesting trials, and gather feedback. Key metrics from such studies include **time saved, reduction in cognitive workload (maybe via NASA-TLX surveys), and user trust scores**. These metrics capture workflow integration and human factors, which are as important as pure accuracy. If an AI is accurate but too cumbersome or confusing, clinicians won't use it. Measuring *Net Promoter Score* or satisfaction of clinicians in pilot usage can predict whether the AI will be adopted at scale.
- **Prospective Pilot Studies (Pre-deployment in limited setting):** Especially for high-impact AI, conducting a prospective pilot in a limited real-world setting before wider rollout is valuable. This is akin to a phase 4 trial but done proactively. For example, test an AI in one unit or one hospital for a period, with oversight, gather outcome data and iterate. While technically this is "deployment," it's part of evaluation before scaling up. Promising methods include A/B testing or stepped-wedge trial designs where some clinicians or sites use the AI and others serve as control, then outcomes (clinical outcomes or process measures) are compared.

Promising Post-Deployment Evaluation and Monitoring:

- **Continuous Performance Monitoring and Feedback Loops:** Once an AI is live, it should not be a static "fire-and-forget" tool. A strong method is implementing ongoing monitoring of the AI's outputs versus actual outcomes. For instance, if our AI predicts a patient is eligible for a trial, we track whether that patient actually enrolls and benefits; if not, why not? This helps catch any drift in the AI's effectiveness. In hospital settings, AI systems often have dashboards tracking their performance (e.g., how many sepsis alerts were true positives vs false). Monitoring metrics could include **real-world accuracy, error rates, override rates (how often clinicians disagree with or ignore the AI), and outcome impact (did using the AI correlate with better patient outcomes or lower costs?)**. For example, Cleveland Clinic monitored its AI sepsis alerts and found a **ten-fold reduction in false positives and a 46% increase in sepsis cases identified** after implementing a particular AI, which validated the tool's performance in practice^[34]. We also look at disparities post-deployment: are there any biases emerging (e.g., is the AI systematically less effective for certain subpopulations)? Continual auditing for fairness is a key part of post-deployment evaluation.
- **Robustness Testing in Production:** Real-world can throw curveballs that weren't in the training data. A good practice is periodic robustness testing of AI in production by injecting test cases. For example, an institution might occasionally run a set of test patient profiles through their live clinical AI (without affecting actual care decisions) to see if outputs remain consistent and robust. Another aspect is monitoring for **data drift**: if the patient population or medical practice changes over time, the AI's model might need recalibration. Automated drift detection algorithms can alert if the input data distribution shifts significantly or if outcome metrics deteriorate. This triggers model retraining or revision. At Massive Bio, we retrain our matching algorithms regularly as new trials and guidelines emerge; we also use **Radiant Core**, our AI control plane, to supervise agents and flag anomalies in the workflow in real time^[13]. This kind of built-in monitoring (agents overseeing agents) is an emerging best practice in AI operations (AI Ops).
- **User Feedback and Governance Committees:** Human-centered post-deployment evaluation includes formalizing user feedback channels. Many hospitals have set up AI or Clinical Decision Support oversight committees that review AI performance and clinician feedback, somewhat like a Pharmacy & Therapeutics committee for algorithms. They track if clinicians are reporting any issues, confusion, or near-misses involving the AI. Metrics like **"number of incident reports related to AI per 1,000 uses"** can be useful. HHS could encourage healthcare organizations to adopt such governance structures, ensuring that AI tools are continually evaluated in context and not just set loose.

Workflow and Human-Centered Evaluation Methods:

Beyond technical performance, evaluating how AI fits into clinical workflow and impacts humans (providers and patients) is vital:

- **Usability Testing:** Before and after deployment, performing usability tests (with methods like heuristic evaluation or direct observation) ensures the AI's user interface and integration are optimal. For instance, does the alert come at the right time in the workflow? Is the information too much, too little? A great AI that is hard to use will fail. Metrics include error rates in using the interface, time to complete tasks, and user satisfaction ratings.
- **Impact on Provider Workload and Burnout:** One promising evaluation angle is measuring how AI affects provider workload. Early examples show AI scribes reduced physician after-hours documentation time, correlating with a **40% drop in burnout in a pilot at Mass General Brigham**[\[35\]](#). On the flip side, poorly designed AI can increase frustration. Therefore, evaluation should include surveys or analysis of provider workload before vs after AI. HHS might fund studies on AI's impact on provider well-being as part of evaluation. If AI can truly reduce burden (a key HHS goal[\[2\]](#)), it should be quantified.
- **Patient-Centered Evaluation:** If the AI directly or indirectly touches patients, we should evaluate patient experience and outcomes. Do patients trust the AI's involvement? Are they more satisfied because decisions are faster or more personalized? For example, in Massive Bio's case, when patients use our AI-powered trial matching through ACS ACTS, we collect feedback on whether it improved their understanding and options. Metrics like patient satisfaction scores, uptake rates of AI-suggested options, and outcomes (like time to treatment or adherence rates) are important. In cancer care, we could evaluate if AI-driven trial matching increases the percentage of patients who get onto a trial or a targeted therapy, and if that in turn improves survival or quality of life (longer-term metrics).

Should HHS support these evaluation processes? Absolutely, and with urgency. Given the rapid proliferation of AI, we cannot rely on ad-hoc evaluation if we want public trust and effective adoption. HHS can play a pivotal role by facilitating systematic evaluation in the following ways:

- **Funding and Grants:** HHS (through AHRQ, NIH, ONC, etc.) should fund research and development of AI evaluation frameworks and tools. For pre-deployment evaluation, grants could support creation of **open reference datasets and bias testing suites** that vendors can use to self-evaluate. For post-deployment, AHRQ could fund projects that develop best practices for continuous monitoring or that measure real-world impact of AI on outcomes. For example, funding a multi-center trial to test how an AI referral system affects cancer patient outcomes across different communities would provide valuable evidence (and inform CMS reimbursement decisions). Specific grant programs (like SBIR for small businesses or ONC's Leading Edge Acceleration Projects - LEAP) could solicit proposals on innovative AI evaluation methods, such as automated validation tools or explainability techniques.
- **Contracts and Cooperative Agreements:** HHS could use contracts to engage entities to perform evaluations on a broader scale. For instance, ONC could contract an academic center or standards organization to develop an "AI in Clinical Care Accreditation" program or voluntary certification (akin to how ONC contracted for development of EHR test procedures). Cooperative agreements could establish centers that partner with HHS to evaluate AI implementations in real healthcare settings (similar to patient safety organizations, but for AI). One idea is a **National AI Testing Collaborative**: HHS sponsors a cooperative network of health systems that agree to pilot and evaluate AI solutions to generate public data on what works and what doesn't. This network can publish results that everyone can learn from, rather than each hospital reinventing the wheel in private.
- **Prize Competitions and Challenges:** As the RFI suggests, prize competitions are a flexible tool. HHS could run an "AI Evaluation Challenge" – inviting proposals for novel metrics or methods to test AI's safety and efficacy, with prize funding to the best ideas[\[28\]](#). Another possibility: an "AI Stress Test Challenge" where participants must design a battery of tests to rigorously probe an AI system for edge-case failures. These competitions can spur creativity and bring in multidisciplinary talent (e.g., tech, human factors, ethics experts) to tackle evaluation.
- **Standards and Frameworks Endorsement:** HHS should collaborate with standards-developing organizations (like IEEE, ISO, or NIST). NIST, for example, has released an AI Risk Management Framework recently; HHS can adopt and tailor this to healthcare, encouraging providers and vendors to use it as a guideline for evaluating and managing AI risks. By referencing such frameworks in HHS communications or even incorporating them into ONC's certification criteria or CMS's expectations for ACOs/hospitals, it nudges the whole industry to systematically evaluate AI. ONC could, for instance, include criteria for "baseline algorithm validation and ongoing monitoring" in its voluntary health IT certification for AI tools, which would essentially operationalize evaluation standards.



In summary, **robust evaluation methods exist or are emerging** – retrospective validation, prospective pilots, continuous monitoring, bias and robustness testing, usability and workflow assessment – but they need to become standard practice. HHS should absolutely support these processes, because without evaluation, we risk patient safety and loss of trust. The mechanisms should be diverse: *grants* to stimulate research, *contracts/cooperative agreements* to implement evaluation programs at scale, and *prize competitions* to crowdsource the best ideas. HHS can also facilitate public-private data sharing that makes evaluation easier (for example, maintain a repository of de-identified patient cases for algorithm testing).

Massive Bio welcomes HHS's support in this area – as an AI company, having clear benchmarks and evaluation protocols helps us ensure our solutions meet the highest standards. In fact, Massive Bio's internal quality system for our AI (with Radiant Core overseeing knowledge, quality, compliance^[13]) could serve as one model for industry. If HHS were to, say, fund a project to pilot a "clinical AI audit toolkit," we would be eager to contribute our expertise. Ultimately, systematic evaluation is what separates hype from reality in AI. By investing in these processes, HHS will foster **trustworthy AI** that providers can rely on and patients can embrace, accelerating adoption in a safe, effective manner.

Question 5. Supporting Private-Sector AI through Accreditation, Certification, and Credentialing – Role of HHS

Q5: "How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?"

HHS can play a catalytic role in empowering the private sector to self-organize and self-regulate in ways that promote safe and effective AI. The key is for HHS to **endorse and reinforce industry-led efforts** by providing guidance, recognition, and where appropriate, incentives or requirements tied to accreditation and certification. Here are concrete ways HHS can support these private-sector activities:

- **Establish an "AI in Healthcare" Voluntary Certification Program:** Just as ONC certifies electronic health record systems, HHS (via ONC or a designated body) could set up a voluntary certification for AI tools used in clinical care. The private sector can drive the criteria development – for example, working through a consortium like the Coalition for Health AI or IEEE – but HHS's support would give it weight. Certification criteria might include core requirements such as: demonstrated accuracy above a threshold, adherence to privacy/security standards, bias testing results, interoperability capabilities (e.g., can ingest/produce data via FHIR), and usability evidence. HHS can support this by convening stakeholders to define the standards and by officially recognizing certified products (perhaps maintaining a public list on HealthIT.gov of "HHS-recognized Certified Clinical AI Tools"). Although voluntary, this signals to healthcare providers which products have met quality bars, accelerating adoption. Moreover, HHS could tie incentives to certification: for instance, CMS could provide a slight increase in reimbursement or a bonus under MIPS for providers who utilize certified AI tools (analogous to how using certified EHR technology was required for incentive payments). Massive Bio sees value in this – as a company, we would pursue such certification to differentiate our AI platform as trustworthy and effective.
- **Support Development of Accreditation Standards for AI Use by Healthcare Organizations:** The private sector, through bodies like The Joint Commission or URAC, could develop accreditation criteria around AI deployment in hospitals and clinics. These might cover governance (does the organization have an AI oversight committee?), training (are clinicians trained in AI tools?), monitoring (does the hospital monitor AI outcomes and safety incidents?), and patient consent/communication about AI. HHS can encourage this by collaborating with these accrediting organizations. For example, HHS could fund a project with The Joint Commission to develop new standards or provide pilot testing opportunities. Once developed, HHS could endorse these accreditation standards as best practice. Eventually, CMS might even incorporate some into Conditions of Participation, but even before that, simply having an accreditation available sets a bar for health systems to strive for. An analogy is how cybersecurity accreditation (like HITRUST) became a de facto requirement in healthcare IT; similarly, an AI governance accreditation could become a norm. HHS's role is to validate and promote that these private standards serve public goals (safety, efficacy, equity).
- **Facilitate Industry-Driven Testing and Benchmarking Consortia:** Private companies often hesitate to share data or methods for competitive reasons, but HHS can provide a neutral ground for collaboration on testing. For instance, HHS could sponsor an **AI Clinical Validation Lab** where multiple AI vendors' tools are evaluated on a common dataset or against common clinical scenarios. This lab could be run by a trusted third party (maybe NIST or a university) under contract with HHS. Participation could be voluntary but if HHS provides reports or "seals of quality" to top performers, it incentivizes companies to participate and improve. The **industry-driven** aspect means the metrics and test cases are decided in consultation with vendors and providers so they're meaningful. We've seen some industry coalitions (like the xVIEW challenges in medical imaging) – HHS can amplify these by lending support and possibly aligning federal needs with them (e.g., linking a challenge to a federal prize or incorporating results into procurement preferences).
- **Credentialing and Training of Healthcare Workforce in AI:** HHS can encourage the private sector in credentialing clinicians for AI competencies. For example, professional boards (like the American Board of Radiology or Oncology

societies) might develop certification modules for practitioners to be “AI-proficient” in their field. HHS can support grants to medical specialty societies to create continuing education and credentialing programs on AI. The private sector (medical boards, specialty colleges) will lead the content, but HHS’s imprimatur or funding can accelerate it. One idea is an “**AI Skills Consortium**” with representation from major medical boards that, with HHS support, defines core competencies for clinicians using AI and develops training materials. If such credentials exist, healthcare organizations could require or prefer that clinicians attain them to use certain high-stakes AI tools, ensuring effective use. For instance, if an AI is used to assist in ICU care, hospitals might want ICU doctors to complete an accredited course on AI in critical care. HHS can further encourage this by linking it to quality programs (e.g., hospitals reporting the percentage of staff certified in AI usage could earn quality credits).

- **Recognize and Promote Exemplars through Public-Private Partnerships:** HHS should continue to partner with leading private entities to showcase success stories of AI implementation (similar to how HHS partnered via the CancerX initiative, which Massive Bio participates in [\[29\]\[30\]](#)). By highlighting case studies of accredited/certified AI tools improving care, HHS builds momentum. Perhaps an annual “**HHS AI in Healthcare Awards**” could be established to recognize healthcare organizations and AI developers that achieve outstanding results (in safety, outcomes, equity). This kind of recognition program, while not directly accreditation, motivates private actors to strive for excellence and gives them a platform to share best practices.
- **Leverage Existing Quality and Safety Programs:** HHS can integrate AI considerations into existing private-sector-led quality programs. For example, Patient Safety Organizations (PSOs) could be encouraged (through AHRQ guidance) to collect and analyze AI-related safety events. HHS could also coordinate with malpractice insurers and organizations like ECRI Institute to develop risk management guidelines for AI – these bodies often create accreditation or approval programs for technologies from a safety standpoint. HHS’s convening power can help unify these efforts so that a coherent set of expectations emerges rather than a patchwork.
- **Support Interoperability and Standards Development:** A form of certification/industry testing that HHS has long supported is in interoperability (HL7, IHE connectathons, etc.). HHS should continue this in the AI context, perhaps sponsoring “plug-a-thon” events for AI integration, where EHR vendors, AI companies, and providers work on integrating AI using standards (like CDS Hooks, FHIR resources for AI outputs, etc.). By doing so, the private sector tools become more plug-and-play, which promotes adoption. HHS can require in its own procurement (for VA, DoD, or federally qualified health centers) that AI solutions support certain interoperability standards, pushing vendors to certify compliance with those standards. This is similar to ONC requiring certified EHRs to support FHIR APIs – which unleashed private sector innovation on top of EHR data. Now we might say certified AI should expose a FHIR-based service for integration, etc.

In essence, **private sector leadership is essential** – they have the expertise and agility to set meaningful standards – but HHS’s backing ensures those standards have legitimacy and reach. The question asks how HHS can best *support* these activities, and we think the answer lies in a combination of **facilitation, endorsement, and incentive alignment**. HHS doesn’t need to control these processes; rather, it should act as a partner.

For example, Massive Bio as a private company is part of multiple collaborative efforts (we mentioned CancerX, and we also work with the Precision Cancer Consortium and other industry groups) aiming to improve data sharing and AI use in oncology. If HHS provides even light-touch support – such as sending representatives to participate, providing grant funding, or officially recognizing outputs of these consortia – it amplifies our impact.

One concrete suggestion: HHS could create an “**AI Clinician User Certification**” in partnership with medical societies (credentialing clinicians in AI use) and an “**AI Tool Certification**” in partnership with standards bodies (credentialing the tools themselves). HHS’s seal on both sides of that equation would greatly accelerate trust and uptake.

Another area is **industry-driven testing**: HHS could coordinate with big health systems and AI vendors to create a **National AI Testing Sandbox**, as mentioned, where AI developers can safely test their algorithms on federated data. Private sector provides the data and tech, HHS provides convening and legal framework (maybe via a DUA or a public-private data trust). This gets around data silos for evaluation.

In summary, HHS should harness the energy and innovation of private entities by providing frameworks and recognition. Think of it as HHS setting the stage and providing the spotlight, while the private sector performers put on the show of developing robust AI and proving it works. This collaborative approach will promote innovation *and* ensure effectiveness, because private stakeholders will buy into the standards they helped create. Ultimately, by supporting accreditation, certification, and credentialing, HHS helps build an **ecosystem of trust and quality** around AI in clinical care – which is the bedrock for sustainable adoption.

Question 6. Successes and Shortcomings of AI in Clinical Care; High-Potential Novel AI Tools

Q6: “Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?”

Successes – AI Tools that Met or Exceeded Expectations: In several domains, AI has already delivered notable successes, improving performance or reducing costs beyond what was expected:

- **Medical Imaging and Diagnostics:** AI for image analysis has shone as an early success. For example, in radiology, FDA-cleared AI algorithms for detecting diabetic retinopathy or identifying pulmonary nodules on CT scans have demonstrated accuracy on par with expert physicians. Many hospitals report that AI-driven diagnostic tools are speeding up workflow and catching findings that might be missed. A concrete success story is in **sepsis detection**: Cleveland Clinic deployed an AI-based sepsis early warning system (Bayesian Health’s platform) and observed dramatic improvements – a **10-fold reduction in false alarm rates and a 46% increase in actual sepsis cases identified in time**[\[34\]](#). The AI alerted clinicians to sepsis in many cases even before antibiotics were given, allowing earlier life-saving intervention[\[34\]](#). This exceeded expectations by both improving clinical outcomes (sepsis detection and treatment timing) and reducing alert fatigue (fewer false positives than prior systems). Another diagnostic success is in pathology: AI tools that analyze digital pathology slides have improved detection of metastatic cancer cells and can do in minutes what might take a pathologist much longer. These tools augment the specialists, leading to faster results and potentially cost savings through efficiency.
- **Operational Efficiency and Cost Savings:** AI has made significant strides in hospital operations. Predictive algorithms for patient flow and staffing have helped some hospitals optimize bed management and reduce length of stay. For instance, AI-driven scheduling tools have cut wait times in operating rooms and imaging departments by predicting no-shows or delays. On a system level, industry analyses project major cost savings; one estimate suggests AI could reduce hospital operating costs by 10–20% in the long term, translating to up to **\$300–\$900 billion in annual savings by 2050** if scaled nationally[\[36\]](#). While much of that is projection, some organizations are already seeing substantial returns on specific AI deployments (e.g., reducing unnecessary ER visits through AI triage, saving millions annually for large health systems). We have even early examples in revenue cycle management: AI for automated medical coding and billing has sped up reimbursements and lowered administrative costs for many practices, meeting expectations for ROI in that area.
- **Administrative and Documentation Burden Reduction:** AI “scribes” and natural language processing tools have been quite successful in reducing clinician burden. Pilot programs with ambient listening devices that transcribe and structure clinical notes have shown that doctors spend less time typing and more time with patients. As noted, **Mass General Brigham reported about a 40% relative drop in physician burnout during an AI scribe pilot**, as it significantly cut after-hours documentation[\[35\]](#). While the financial ROI for scribes is still being assessed, the improved provider satisfaction and reclaimed time have exceeded expectations in those trials – a critical outcome given the burnout crisis.
- **Clinical Decision Support and Personalized Medicine:** In specialized areas like oncology, AI-driven decision support has started to exceed expectations by handling complexity beyond human limits. Massive Bio’s experience provides an example: our AI-driven trial matching “pre-screening hubs” have **dramatically shortened the time to identify clinical trial options for cancer patients and boosted trial participation rates**, as recognized by the White House Cancer Moonshot[\[12\]](#). These hubs scan comprehensive genomic profiles and patient data to pinpoint trials in real-time, achieving in minutes what used to take expert teams many hours. The result is more patients getting matched to trials (over 50,000 patients per year projected) and faster enrollment – outcomes that exceed prior benchmarks and expectations for expanding trial access[\[11\]\[12\]](#). This has also shown cost benefits: every patient who goes on a trial often gets expensive therapy covered by the trial sponsor, potentially saving payers tens of thousands of dollars per patient compared to standard care.
- **Population Health and Preventive Care:** AI risk prediction models have helped focus interventions on the right patients. For example, some ACOs use AI to predict which patients are at risk of hospitalization in the next 30 days and then proactively deploy care managers to those patients. In some cases, these models have exceeded expectations by identifying high-risk individuals that traditional methods missed, leading to measurable reductions in readmissions or ER visits. Insurers, too, have used AI to flag potential medication non-adherence or care gaps and have improved outcomes with relatively low-cost outreach, saving larger costs down the line.

Shortcomings – Where AI Tools Have Fallen Short:

- **Generalizability and Real-World Performance Gaps:** A number of AI tools that performed well in research settings failed to meet expectations in the real world. One famous example is IBM Watson for Oncology – heavily touted as an AI to

revolutionize cancer treatment, it ultimately fell short of clinical expectations and was not shown to improve patient outcomes meaningfully. This highlighted that domain-specific nuance and integration into workflow were underestimated. In some cases, AI algorithms have shown **degraded accuracy when deployed on different patient populations or different hospitals** than they were trained on, exposing a lack of generalizability. For instance, an AI trained on one hospital's imaging data might not perform as well on another's due to different equipment or patient demographics. These disappointments have made clear that rigorous external validation is needed; some early AI products did not get that, leading to over-promising and under-delivering.

- **Bias and Health Equity Issues:** A notable shortcoming was uncovered in an algorithm widely used to prioritize patients for care management in some health systems: it systematically underestimated the health needs of Black patients compared to white patients because it used healthcare spending as a proxy for illness, and historically less money was spent on Black patients[22]. This bias meant the tool fell short morally and practically by missing high-need patients. It was an eye-opener that some AI can inadvertently perpetuate disparities – a failure to meet the expectation that AI would be objective and fair. HHS has rightly flagged this as an area needing attention. Patients and providers expect AI to at least do no harm in terms of equity, and when those expectations are violated, it undermines trust.
- **Lack of Clear Financial ROI in Some Cases:** While AI can save costs, in some deployments the ROI has been elusive or slower than expected. For example, early adopters of AI for clinical documentation found that while doctors were happier, the financial ROI wasn't immediate because these tools can be expensive to implement and maintain[37]. Some hospital executives expected quick cost savings from AI, but realized that benefits sometimes accrue in intangible ways (quality, patient satisfaction) or long-term (averted costs in the future) rather than immediate budget relief. As noted in a FierceHealthcare evaluation, AI scribes improved experience but had "limited financial ROI" initially[37]. This shortfall in meeting cost expectations is often due to integration costs, training, and the need to adjust workflows. Similarly, some predictive models didn't yield savings because organizations struggled to act on the predictions (the "last mile" problem). This taught us that AI alone doesn't save money; it must be paired with effective process changes, and that piece sometimes fell short.
- **Workflow Integration and Alert Fatigue:** Some AI tools under-delivered because they weren't well integrated or produced too many alerts. For example, early AI-based clinical decision support for EHRs sometimes generated excessive suggestions or warnings that annoyed clinicians (e.g., constantly reminding about care gaps or suggesting irrelevant clinical trials). This created alert fatigue and low utilization, falling short of the hoped-for impact on quality. The issue wasn't that the AI was "wrong" – rather it wasn't nuanced or user-friendly enough, betraying expectations that AI would *simplify* workflows when in fact it sometimes complicated them. The lesson learned is the importance of human-centered design, which some early solutions lacked.

In summary, AI has excelled in well-defined tasks like image analysis, prediction of specific events like sepsis, and automating routine processes – often exceeding expectations in speed and accuracy. It's faltered when faced with issues of generalization, bias, user adoption, and translating predictions into practice. Many shortcomings have been due to **context, not core algorithmic ability**: integration, trust, data quality, etc.

High-Potential Novel AI Tools for the Future:

Looking forward, we see several categories of novel AI tools that could have transformative impacts on outcomes, quality insights, and cost reduction:

- **AI for Precision Medicine & Complex Decision Support*: *AI that can synthesize multimodal data** (genomics, proteomics, imaging, clinical records) to guide personalized treatment stands to greatly improve outcomes, especially in complex diseases like cancer, cardiac disease, or neurological conditions. The next generation of decision support might use deep learning to identify hidden patterns – for instance, predicting which cancer patients will respond to immunotherapy by integrating genomic markers with pathology images and clinical history. Such an AI could spare patients ineffective treatments (improving outcomes and avoiding costs of toxicity). It could also suggest novel uses for existing drugs (drug repurposing) or optimal trial opportunities. The potential to give new insights on quality comes from uncovering subgroups of patients who have vastly different outcomes with different approaches – something traditional analysis often misses. Massive Bio's work hints at this potential: by analyzing real-world data across thousands of patients, our AI has started to identify patterns (like particular tumor mutations combined with patient demographics) that correlate with better trial responses. In the future, a clinician might get an AI-generated "patient similarity" analysis: "patients like this one, with X and Y characteristics, tended to live 6 months longer on treatment A vs treatment B."** This kind of guidance could revolutionize quality of care by truly individualizing it.
- **Generative AI and Large Language Models (LLMs) in Clinical Care:** The emergence of powerful LLMs (akin to ChatGPT) trained on medical knowledge has huge potential. Novel tools using LLMs could function as **AI co-pilots** for clinicians –

e.g., synthesizing a patient’s entire chart and providing a concise summary with key recommendations, or even drafting clinical plans that the doctor can refine. This can save time (reducing costs associated with documentation and information retrieval) and potentially improve quality by ensuring no detail is overlooked (the AI can cross-check treatments against guidelines and patient specifics). Careful deployment of LLMs could also improve patient communication: imagine an AI agent that converses with patients to gather history or answer common questions with empathy and clarity, extending the care team’s capacity. These tools are novel and need vetting, but their ability to understand and generate human-like medical dialogue is promising for addressing two needs: freeing up clinicians and engaging patients better. If implemented with guardrails, they could reduce administrative burden dramatically (cost saving) and enhance patient satisfaction and understanding (quality).

- **Predictive Analytics for Preventive and Value-Based Care:** AI that predicts events like hospital admissions, disease progression, or complications can enable preemptive action. A novel wave of predictive tools, improved by AI, could manage chronic diseases far more effectively – for example, AI monitoring of heart failure patients via wearables, predicting acute decompensation days in advance so that outpatient intervention prevents a costly hospitalization. Such tools would reduce costs (by avoiding hospital stays) and improve outcomes (by preventing crises). Another high-potential area is AI for **frailty and dementia**, conditions highlighted by HHS for coming decades^[38]. New AI models using data from daily activities, sensors, and routine health info could flag early cognitive decline or frailty risk, prompting earlier therapy or home modifications that maintain quality of life and reduce later institutionalization costs. HHS focusing research on these would be wise (ties into Q10).
- **Automation of Routine Tasks and Robotic Process Automation (RPA):** While not as glamorous as clinical diagnostics, AI-driven automation of repetitive administrative tasks holds massive cost-saving potential. From appointment scheduling bots to AI handling prior authorization paperwork (retrieving required info, populating forms) to automated image post-processing, these novel applications free human staff to do higher-level work. The savings from reducing administrative overhead (which is huge in U.S. healthcare) could be significant. For instance, an AI that processes claims or billing might drastically cut billing errors and time, or an AI that manages inventory and supply chain in a hospital could prevent stock-outs of critical supplies or overstock of expensive items.
- **Decentralized and At-Home Care via AI:** Tools that allow hospital-level monitoring and treatment at home are emerging. AI can interpret data from home devices (like oxygen saturation monitors, glucometers, etc.) to manage patients remotely. Novel AI that integrates into telehealth platforms could triage which home patients need in-person visits, which can be managed virtually, optimizing resource use. This area is ripe to both cut costs (hospital at home is cheaper than hospital in-building) and maintain or improve outcomes (patients often prefer to heal at home, lowering infection risk too). An example might be an AI that monitors a post-surgery patient’s wearable data and virtual check-ins to detect early signs of complication and arrange a home nurse visit immediately – heading off an ED visit.
- **Clinical Trial Optimization and Drug Development:** AI tools can supercharge R&D, indirectly improving clinical care by getting better therapies to market faster and cheaper. AI in trial design (simulating trials, optimizing patient selection criteria) and in pharmacovigilance (monitoring for drug side effects in real time) are novel areas. If trials become more efficient, costs of bringing new treatments down might drop, eventually reducing drug costs for the system. For example, AI-driven **digital twins** (virtual patient models) could be used to predict outcomes, reducing the size or number of real trials needed. This might be a bit farther horizon, but high potential.

In summary, the greatest potential lies in **AI that amplifies human capabilities in complex decision-making and personalizes care**, as well as AI that **relentlessly drives efficiency in the background**. The goal is a health system that is smarter (insights we didn’t have before), more precise (right treatment for right patient at right time), and leaner (less waste and friction). Achieving this could dramatically improve outcomes and cut waste – the two pillars of value in healthcare.

From Massive Bio’s vantage point, we are especially excited about AI’s potential in **democratizing clinical research and precision oncology**. By using AI to connect diverse patients with the best therapy options (standard or experimental) regardless of where they live or who they are, we can improve outcomes and reduce inequities. That also reduces cost in a disease like cancer – by avoiding ineffective therapies and focusing on those likely to work, or by utilizing clinical trials (often funded by sponsors) instead of expensive commercial drugs when appropriate. For instance, if a \$150,000/year cancer drug might not work for a patient due to their tumor profile, but an AI identifies a trial with a promising agent available to that patient, and that agent turns out effective, we’ve potentially saved six figures and improved the patient’s health.

Conversely, we caution that continued attention is needed to avoid repeating past shortfalls. Each high-potential AI tool must be developed with robustness, fairness, and workflow in mind to truly deliver. HHS’s enabling policies can help ensure these novel tools reach their potential by guiding development toward areas of genuine need (like dementia, frailty, health disparities) and by smoothing the path for those that prove their value. The successes so far give us confidence that with the right support, the next wave of AI tools will indeed meet or exceed our expectations – perhaps even achieving the

“transformative force” across healthcare that leadership envisions[39].

Question 7. Influential Roles in AI Adoption and Primary Hurdles in Health Care Organizations

Q7: “Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?”

In our experience working with a variety of healthcare organizations (from major academic medical centers to community oncology clinics), certain key decision-makers and bodies heavily influence whether AI tools are embraced or not. Additionally, there are common administrative hurdles that organizations face when implementing AI. Identifying these roles and hurdles is important because it helps target interventions (who to educate, what processes to streamline) to speed up AI adoption.

Key Roles and Decision Makers Influencing AI Adoption:

- **Executive Leadership (C-Suite):** Hospital and health system executives – particularly the CEO, Chief Medical Officer (CMO), Chief Information Officer (CIO), and Chief Technology/Innovation Officer – play a decisive role. If the C-suite believes in a strong digital/AI strategy, the institution will pursue AI projects and allocate resources. Conversely, if top leadership is skeptical or risk-averse, AI initiatives often stall. The **Chief Financial Officer (CFO)** is influential from a budget perspective – their view on ROI can make or break an AI investment. Many successful AI adopters have had CEOs who champion technology and CIOs/CTOs who have developed an AI roadmap. We’ve observed that large integrated systems with visionary leadership (some even creating roles like Chief Data Scientist or VP of AI) are far ahead in adoption rates, whereas organizations without that leadership focus lag behind[21].
- **Clinical Champions (Physician/Nurse Leaders):** Within clinical departments, respected physicians or nurse leaders who advocate for an AI tool can significantly sway their peers and department’s willingness to adopt. For example, a Chair of Radiology who is pro-AI can push their department to adopt AI in imaging faster, working through kinks and getting buy-in. Many hospitals have a **Chief Medical Information Officer (CMIO)** or similar role – often a physician – who bridges IT and clinical practice; these individuals are critical in evaluating AI tools and persuading other clinicians of their value. If the CMIO or a departmental chief says “this AI will help us and I’m on board,” that endorsement carries weight. Conversely, if frontline clinicians are not involved or are opposed, no amount of executive push will succeed – they can refuse to use the tool or undermine it. Thus, identifying and empowering *clinical champions* is key. In our Massive Bio implementations at cancer centers, we always identify an oncologist or research nurse leader who becomes the internal champion for our trial-matching AI; their influence helps integrate the tool into the care flow.
- **Governance Committees (e.g., IT Governance, Value Analysis, Ethics Boards):** Many health organizations have formal committees that evaluate new technologies or practices. These can include a **Value Analysis Committee** (which reviews new clinical products for efficacy and cost), an **IT Governance Board**, and sometimes specialized **AI or Digital Health Committees** in more advanced systems. These bodies typically include a mix of administrators, clinicians, IT staff, and compliance officers. They influence adoption by scrutinizing proposals for new AI tools – looking at evidence, cost, alignment with strategic goals – and then approving or denying them for purchase or pilot. If such a body is progressive and educated about AI, it can facilitate adoption by giving the green light and funding. But if it’s overly conservative or unfamiliar with evaluating AI, it can be a bottleneck. We’ve seen cases where a promising AI solution gets stuck for a year in committee deliberations over data security or uncertain clinical benefits. Strengthening these governance processes (with clear criteria for evaluating AI, perhaps training members on AI basics) can help.
- **IT Department / Chief Information Security Officer (CISO):** The IT team, including security and infrastructure personnel, heavily influence whether an AI solution can be implemented. They evaluate integration effort, cybersecurity risks, and compatibility. A supportive CIO/CTO will allocate resources to integrate an AI system with the EHR or data warehouse; a cautious one might cite concerns and delay integration. The CISO’s approval is often needed if an AI involves cloud or external data sharing. We often see great ideas slowed by months due to data security reviews and legal contracting for data use. So while these roles might not “decide” on the business need, they have veto power if risk is deemed too high or if the project demands IT work they can’t support.
- **Payers and External Decision Makers:** Within provider organizations, the above are primary. But we should mention that external actors like **insurance payers** or regulators can indirectly influence adoption. For example, if a payer starts reimbursing an AI-based service, the hospital CFO is more likely to approve that project. If a regulator or accreditation body starts requiring AI-related competencies, the CMO will push to implement AI to stay compliant. Thus, while not inside the organization, these external forces change the internal calculus of decision makers.

Primary Administrative Hurdles to AI Adoption:

- **Budgeting and ROI Justification:** One of the biggest hurdles is simply getting budget approval. Hospital budgets are tight, and AI tools (plus the requisite IT infrastructure) cost money upfront. Administrators ask: *What's the return on investment?* Because many AI benefits are indirect or long-term (improved outcomes, operational efficiencies that are hard to quantify immediately), making the business case is challenging. Often there is no line item for "AI" in budgets – it competes with other urgent needs (new MRI machines, hiring staff, etc.). The uncertainty around ROI leads to slow or no adoption. This is an administrative barrier because even if clinicians want it, the finance department might say no. We need more demonstration of ROI (or at least value) to overcome this, as well as novel financing models (like vendor risk-sharing or phased pilots). It would help if, as discussed earlier, reimbursement or savings from AI could be captured more directly – that would ease CFO concerns.
- **Procurement and Contracting Processes:** Hospitals have complex procurement processes that can be a hurdle. Any new AI vendor has to go through legal contracting, often addressing data use, IP, liability, etc., which can be arduous especially if the contracts office is not versed in AI-specific issues. Things like intellectual property ownership of models or data sharing rights can cause delays. Additionally, some organizations require competitive bidding or extensive due diligence for new vendors, which prolongs adoption. Many AI companies are startups without decades of track record, which makes procurement committees nervous. We've seen several month lags in just negotiating data use agreements. Streamlining these processes (perhaps with template AI contracts, as recommended earlier) is needed.
- **Integration with Existing Systems (Technical Hurdle):** Administratively, coordinating between the AI vendor and internal IT for integration can be a hurdle. Schedules, priorities, and technical resources must align. If an AI can't plug easily into the EHR or workflow, it may end up as another separate system clinicians have to log into – which is a barrier to adoption (clinicians resist multiple logins/screens). So from an admin perspective, the hurdle is marshaling the needed IT changes or interfaces, which might require approvals from EHR vendors or internal change control processes. This is time-consuming and sometimes costly (EHR vendors might charge for integration work). Until interoperability improves, each new tool has this hurdle. **Interoperability improvements** (discussed in Q8) would directly tackle this.
- **Training and Change Management:** Adopting AI isn't just a software install; it's a process change. Administratively, planning and executing training for staff is a hurdle. Hospitals need to schedule training sessions, develop new policies (e.g., how to override AI recommendations, documentation of AI usage), and monitor compliance. Change management is hard in healthcare – people are used to certain routines. Some administrators might be hesitant to introduce yet another change given how burdened clinicians already are. Without strong training and support, an AI tool could flop, so leadership needs to invest in that. But dedicating time for training takes providers away from clinics, which has a cost. Some might not see it as a priority unless mandated.
- **Data Governance and Privacy Concerns:** Many organizations have internal rules about data sharing (like, they may not allow PHI to be sent to cloud services, etc.). Getting approvals through privacy boards or compliance committees can be a hurdle especially if the AI is cloud-based or requires big data transfers. There may be concern about HIPAA compliance, and administrators in compliance/legal roles might press "pause" until thoroughly convinced. While these are legitimate concerns, sometimes the caution is disproportionate to actual risk and slows beneficial projects.
- **Measurement and Proof of Concept Requirements:** Some institutions require extensive pilot studies internally before full adoption – an administrative decision to mitigate risk. This means additional time to do an internal trial, measure outcomes, present results to committees, etc. It's a hurdle because it delays scaling. On the flip side, places that skip this and jump to adoption might face backlash if something goes wrong. So it's a careful balance.
- **Cultural Resistance and Workforce Acceptance:** Though not a formal administrative process, the culture is an underlying factor. If an organization's culture is not innovation-friendly or if staff fear AI (job displacement fears, "not invented here" syndrome), adoption faces an uphill climb. Administrators must manage these cultural factors – e.g., communicating that AI is there to assist, not replace, and incorporating frontline feedback into selection and implementation. Where culture has been addressed (through inclusive decision-making and transparency), adoption is smoother; where it hasn't, AI tools get silently sidelined by users.

To illustrate, consider a smaller community hospital: The CEO and CFO might be unsure about an expensive AI for ICU monitoring. The ICU director is interested but the value committee says "show us evidence from similar hospitals." IT says integration with their older EHR will be tough. Meanwhile, nurses worry the AI alarms might add workload. All these need addressing by the right leaders: the ICU director championing, the CIO ensuring integration, the CEO aligning it with strategy (like to be a high reliability org), and the CFO perhaps agreeing to a pilot with vendor risk-sharing. If any of these pieces break down, adoption stalls.

Massive Bio often engages with **oncology service line directors or research administrators** at hospitals when offering our AI-matching service. If the director sees how it helps patients and research, they influence the CMO or CEO. But often an admin

hurdle we see is “This sounds great, but do we have manpower to coordinate the patients that get matched? Who will own this process internally?” That is a valid administrative hurdle – responsibility and workflow assignment. If unclear, they hesitate to start. We help by offering concierge support (our staff to supplement theirs), which alleviates the hurdle.

In summary, **the most influential actors** are executive leaders (C-suite), tech and data leaders (CIO/CMIO/CISO), and clinical champions/department heads, often guided by governance committees. **Primary hurdles** are financial justification, complex procurement and integration processes, the effort needed for training and change management, and internal risk/compliance gating. To accelerate AI adoption, we need to **engage and educate those key decision-makers** (for example, demonstrating ROI to CFOs, assuring CIOs on security, showing clinicians evidence of benefit) and **streamline administrative processes** (templates for contracts, better interoperability, etc.). If HHS and industry groups could provide tools to health systems – say, an “AI adoption playbook” for hospital administrators – it could help organizations navigate these hurdles more confidently.

Removing or easing these obstacles will empower those influential leaders to say “yes” to AI with fewer reservations. For instance, if CMS or HHS provides clear evidence or endorsement of an AI solution’s efficacy (like via certification or inclusion in best practice guidelines), it makes it easier for committees and executives to approve it. Knowing that others have succeeded (perhaps through HHS sharing case studies) also gives decision-makers cover to proceed. Ultimately, identifying **who holds the keys** and **what roadblocks exist** allows targeted strategies to unlock AI adoption at the organizational level.

Question 8. Interoperability Opportunities to Fuel AI Development and Widen Adoption

Q8: “Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.”

Interoperability – the seamless exchange of health data – is the oxygen that AI needs to breathe and flourish in clinical care. Enhanced interoperability would have far-reaching benefits: it would enlarge the available data for AI model training (fueling research), make it easier to deploy AI in different settings (widening market opportunities), and enable new types of AI-driven workflows that are currently impractical (accelerating adoption). Below we outline key areas where improved interoperability is most needed and the specific data types/standards involved:

- **Integration of Electronic Health Records (EHRs) with AI Tools:** A top priority is making clinical data in EHRs easily accessible to AI developers and systems via standardized APIs. The advent of **FHIR (Fast Healthcare Interoperability Resources)** is a huge step forward. Enhanced use of FHIR (and full compliance with the 21st Century Cures Act’s interoperability and information blocking rules) will widen opportunities by allowing third-party AI apps to plug into EHRs securely. For example, if a hospital’s EHR exposes a FHIR API for patient demographics, problems, medications, labs, etc., an AI tool can fetch those in real-time to analyze and provide decision support (e.g., suggesting a clinical trial or detecting a care gap) without custom integration. Standard FHIR profiles for specific needs (like scheduling, imaging results, pathology reports) could let AI modules operate across many hospitals with minimal tweaks – this opens the market for smaller innovators to serve multiple sites. ONC’s recently finalized regulations on API standards need to be fully implemented and extended. Specifically, **bulk FHIR data access** (using FHIR Bulk Data / Flat FHIR) is crucial for research and population-level AI. It allows pulling de-identified datasets of EHR records across thousands of patients for model training or quality improvement analysis. If every EHR can export bulk data in a standard way (with proper permissions), researchers won’t have to spend 80% of time cleaning and normalizing data – they can focus on the AI algorithms. HHS should continue pushing bulk data standards and perhaps provide **synthetic or test FHIR datasets** for AI benchmarking (like a realistic synthetic health record database on which AI can be evaluated fairly).
- **Standardization of Key Clinical Data Types (e.g., Genomics, Imaging) for Interoperability:** Certain high-value data types are currently not as interoperable as they could be:
- **Genomic Data:** Precision medicine and AI go hand in hand, but genomic data (like DNA sequencing results, variant interpretations) often reside in PDFs or proprietary lab portals. Adopting standards like HL7 FHIR Genomics or the GA4GH (Global Alliance for Genomics and Health) schemas, and integrating those into EHRs, would greatly help. For instance, the **mCODE (minimal Common Oncology Data Elements)** standard, built on FHIR, defines how to represent oncology-specific data, including genomics, in a structured way^[10]. If genomic results (mutations, biomarkers) are interoperable and readily pulled into AI systems, it fuels advanced oncology AI that can identify trials or therapies matching a patient’s molecular profile. Massive Bio’s platform uses native FHIR/mCODE integration to connect EHRs and labs^[10], which was essential for scaling our AI across different clinics. Widespread adoption of such standards will allow any oncology AI tool to work in any oncology practice system, widening the market dramatically (currently, AI companies have to custom-map data from each provider’s system).
- **Imaging and PACS:** Radiology and pathology images are typically stored in PACS systems using DICOM standards, which is good, but making those images accessible to AI algorithms outside the local network is challenging. Enhanced

interoperability via cloud image exchanges or APIs that can provide images (with appropriate patient consent) to AI model providers would accelerate development of imaging AI – researchers could access diverse imaging data, and hospitals could plug in external AI services more easily. Initiatives like DICOMweb (RESTful APIs for DICOM) should be advanced and linked with FHIR for orders and results. If an AI developer can, with permission, query “give me all chest X-rays and outcomes for patients with COVID across 10 hospitals” through an interoperable network, it massively fuels research. TEFCA (Trusted Exchange Framework and Common Agreement) might be extended in the future to include image exchange, not just documents – that’s an area to explore.

- **Unstructured Clinical Notes:** A lot of important info lives in doctors’ free-text notes. Natural language processing (NLP) AI thrives on this, but currently exchanging notes reliably is tough (CCDA documents carry some notes, but often in blob form). Enhanced interoperability could mean adoption of standards like **C-CDA and FHIR DocumentReference** to share notes in structured formats. Also, using NLP-specific standards like CTAKES annotations or FHIR Communication resources for note content could help. Imagine if a researcher could query a network for “de-identified notes of patients who had symptom X” through a standard service – it would enable training conversational AI or symptom-checker models at scale.
- **Patient-Generated Health Data and Devices:** Data from wearables, home devices, patient-reported outcomes etc., is increasingly valuable for AI (e.g., in chronic disease management). Interoperability here is often overlooked. Standards like IEEE 11073 (for device data) and frameworks to integrate patient-generated data into EHRs (projects like CARIN Alliance for consumer data) should be advanced. If these data streams can flow into the clinical record in standardized ways, AI can analyze continuous monitoring data to predict issues. For example, interoperability standards enabling a patient’s smart watch ECG or glucometer readings to be pulled into an AI arrhythmia or hypoglycemia prediction model across providers would widen telehealth and remote monitoring AI use.
- **Interoperability to Link Clinical and Research Data:** A major opportunity is connecting clinical care data with clinical trial and research data. For instance, **clinical trial registries** (like ClinicalTrials.gov) are a rich source of eligibility criteria and trial availability, but historically it’s been separate from EHR data. If criteria could be standardized (there are efforts to use FHIR for trial eligibility criteria) and patient data can be matched to them systematically, AI can automate trial matching at scale. Massive Bio does this by parsing ClinicalTrials.gov and matching with patient EHR data; the more those formats are standardized, the more efficient and universal this becomes^{[40][7]}. Enhanced interoperability between healthcare and research also means if a patient consents, their EHR data could flow into research databases (e.g., for learning health networks). HHS could encourage adoption of standards like OMOP common data model to allow easier pooling of multi-center data for AI training. Also, consider linking claims data with clinical data – standards to integrate Medicare claims or payer data with EHR data on the fly could allow AI to factor in cost and utilization patterns, which might reveal insights about quality and efficiency (like identifying high-cost patient trajectories that could be improved).
- **Benchmarking Tools and Shared Datasets:** The question also mentions benchmarking tools. Interoperability can enable shared **benchmark datasets** for AI. For example, if multiple health systems can contribute de-identified data to a central repository because they all speak the same data standards language, you can create a national “benchmark dataset” for, say, diabetic patient outcomes, on which AI models from different vendors can be benchmarked. A current challenge is that each AI vendor might test on their own proprietary dataset – not apples-to-apples. HHS could coordinate creation of **reference datasets** (with patient privacy preserved) that are continuously updated via interoperable feeds from diverse sources. These could be used to benchmark algorithm performance fairly. Perhaps the NIH or AHRQ could host such repositories (some exist, like MIMIC for ICU data, but expanding to other domains).
- **Enhanced Interoperability in Social Determinants of Health (SDOH) Data:** Often overlooked, but SDOH (transportation, housing, social support data) is critical for AI to truly understand patient context and improve outcomes. We need standards (HL7 Gravity Project is defining some) to record and share SDOH in health records. If an AI can access standardized SDOH data (like food insecurity status, or transportation needs), it can better tailor interventions and identify at-risk patients. For instance, Massive Bio’s ACS ACTS program not only matches trials but also identifies social needs like transportation and lodging^{[41][42]}. Because ACS has integrated that data, our AI workflow can prioritize solutions (like connecting to ACS’s Road to Recovery transport program^[43]). If all that SDOH information was interoperable, any AI care coordination tool could systematically account for barriers to care – improving real-world effectiveness and equity.

Where enhanced interoperability would specifically widen market opportunities: If data flows more freely (with patient consent and privacy safeguards), the barrier to entry for AI startups lowers – they don’t have to build custom pipelines for every hospital, meaning more competitors and solutions in the market. For example, right now an AI developer might avoid building a tool for small clinics because integrating with their EHR is too costly. With universal FHIR APIs, that market opens up. Enhanced



interoperability also means AI solutions can be EHR-agnostic – a huge selling point. Providers would have more choices (not locked into AI offered by their EHR vendor only). This competition can drive down costs of AI and improve quality of solutions.

How interoperability fuels research: Perhaps the biggest impact – researchers (academic or industry) spend enormous effort on data wrangling. If HHS ensures that more high-quality, diverse datasets are accessible in standardized formats (possibly through a national health data exchange under TECA that includes research as an allowed purpose with proper governance), AI research accelerates. One could imagine a secure national research environment where authorized researchers can run AI training on aggregated EHR data from across the country (with all identifiers removed or using a federated learning approach). That’s transformative – it could lead to discoveries of patterns for rare diseases, or development of AI for underrepresented populations (since currently, many AI models skew toward the data from big academic centers in urban areas[21], leaving rural or minority-heavy populations less represented; broad interoperability would bring in those missing pieces).

In our own operations, we’ve seen how interoperability makes a difference. Massive Bio’s Reticulum Nexus platform is “interoperable by design” – it natively integrates with FHIR and mCODE, and connects EHRs, labs, CROs, etc., which was critical for operating across many partners[10]. This interoperability allowed us to align patients, providers, and sponsors on one data fabric to scale trial enrollment[44]. Without those standards, doing this at a national level would be near impossible. It’s a microcosm of what could happen nationwide if everyone adopts interoperability: you unlock **network effects**. The more data sources connect, the more powerful the AI insights and the broader the impact.

Specific Suggestions: - HHS should continue strong implementation of information blocking rules so that providers and EHR vendors cannot refuse legitimate data sharing requests. Patients should be empowered to direct their data wherever (via apps, etc.) – this patient-mediated interoperability could feed AI-driven patient apps and research. - Invest in **health information exchanges (HIEs)** to incorporate AI support – e.g., HIEs could offer analytics or AI-ready datasets as a service to participants. - Promote or require standard **benchmarks**: For example, ONC could coordinate with NIST to produce a set of synthetic yet realistic patient records (covering diverse scenarios) that any AI for a given use case should be tested against. If these are FHIR formatted and publicly available, any developer can benchmark and any buyer can demand to see results of that benchmark, which in turn pushes developers to train on more interoperable data.

In conclusion, enhanced interoperability – especially via common **data standards (like FHIR, DICOMweb, mCODE)** and **data exchange frameworks (like TECA)** – would supercharge both the **supply side of AI (research and development)** and the **demand side (deployment and adoption)**. It widens market opportunities by lowering integration costs, fuels research by easing data access across silos, and accelerates development by enabling collaborative benchmarking and learning from broader datasets. We cannot overstate the importance of this; indeed, as HHS leadership noted, “AI is powered by data. Data liquidity and the trust in how data moves are essential.”[18]. Massive Bio fully agrees and supports HHS initiatives to make data flow more freely and securely. It’s akin to building the interstate highways for data – once in place, innovation in AI can travel and spread at unprecedented speed, reaching every community and patient with its benefits.

Question 9. Patient and Caregiver Challenges for AI to Address – and Their Concerns about AI Adoption

Q9: “What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?”

From the patient and caregiver perspective, AI is not about algorithms – it’s about tangible improvements in their healthcare experience and outcomes. Massive Bio engages directly with patients (particularly cancer patients and their families) through our services, and we have gleaned insights into what they hope AI can solve and what worries them about AI.

Challenges Patients/Caregivers Want AI to Address:

- **Access to the Right Care and Information:** One of the biggest patient pain points is navigating the healthcare system and finding the best treatment options. Patients want AI to help cut through the complexity. For example, cancer patients often ask, “Is there a clinical trial or a new therapy that could help me?” Many feel overwhelmed trying to research on their own[45]. They would love AI-driven tools that *proactively present them with personalized options*. Our ACS ACTS partnership is an example: patients fill a simple form and our AI matching provides tailored trial options, which patients find empowering[46][45]. Patients desire this kind of **personalized guidance** at critical decision points – which hospital is best for my condition, which treatment path likely fits me, etc. In general, they hope AI can democratize expertise, so that even if they aren’t at a top academic center, they still get cutting-edge recommendations. Caregivers, likewise, want tools that help them help their loved ones – e.g., an AI that could summarize a patient’s condition and list questions to ask the doctor, or predict care needs.

- **Timely and Accurate Diagnoses:** Patients often express frustration about missed or delayed diagnoses. They would like AI to assist in catching problems earlier. For instance, in radiology or pathology, if AI can detect a tumor or abnormality that a busy clinician might overlook, patients see value – it could mean an earlier cancer diagnosis or avoiding a missed finding. Particularly in fields like rare diseases, patients want AI to shorten the diagnostic odyssey by recognizing patterns in symptoms that might point to a condition sooner than a human would.
- **Better Coordinated Care and Reduced Administrative Burdens on Them:** Navigating appointments, refills, referrals – these are pain points patients hope AI can ease. They want the health system to be more coordinated and responsive. An example challenge: a patient might see multiple specialists and feels like they have to retell their story each time and chase after records. They would be thrilled if AI could integrate their information so every provider is up to date (a function of interoperability and AI summarization). Also, tasks like scheduling or medication refills – patients would love intelligent systems that make these seamless (e.g., an AI that knows when you’re due for a check-up or notices you haven’t picked up a prescription and sends a reminder or help). This ties to quality of life; patients often feel they are project managers of their own care – AI could lighten that load by proactively handling some logistics.
- **Improved Treatment Experiences and Outcomes:** Ultimately, patients and caregivers want better health outcomes – longer survival, less pain, improved quality of life. They see AI as a potential means to tailor treatments (minimizing trial-and-error) and to manage chronic conditions better day-to-day. For example, a diabetic patient might want an AI coach that helps keep their blood sugar stable with personalized tips, or an AI that predicts a flare-up of a condition so they can act early. Caregivers of elderly or chronically ill patients might want AI monitoring that alerts them or providers if something is off (like subtle signs of stroke or deterioration in a dementia patient), preventing serious events. Essentially, they hope AI can provide an “early warning system” and continuous support that current periodic visits cannot.
- **Equity in Access:** Patients in underserved areas or from marginalized communities often voice that they lack access to specialists or trials. They hope AI can bridge that gap – e.g., via telemedicine AI or remote monitoring that connects them to top expertise. They also want assurance that treatments are designed for people like them. For instance, minority patients might feel more confident if AI can help ensure the care they receive is based on data that includes people of their background (avoiding one-size-fits-all medicine).

To sum up the challenges: **finding the right care faster, coordinating that care, and personalizing it to improve outcomes** – these are what patients really care about. AI that addresses these will be embraced. For example, ACS ACTS users appreciate that the service helps them “*navigate clinical trials and find the best treatment options available*”, which was previously overwhelming^[47].

Concerns Patients/Caregivers Have about AI:

- **Privacy and Data Security:** A foremost concern is “What happens to my data?” Patients worry that if AI systems are pulling together their health information, it could be misused or fall into the wrong hands. They want to know that their personal details and medical history remain confidential. There’s fear of hacking or leaks. For instance, if an AI is analyzing their genomic data or insurance records, they ask: will this be shared with third parties, could it affect my insurance coverage, or could an employer somehow access it? They need reassurance through transparency and strong privacy protections that AI won’t compromise their confidentiality^[48]. Our experience is that patients are often willing to share data if it helps their care or others, but they want to *actively consent* and have control. So consent and clear communication is key.
- **Loss of Human Touch and Personal Interaction:** Many patients fear healthcare could become impersonal if AI takes on too much. The human connection with doctors and nurses is a source of comfort; there’s concern that AI might replace that or make providers overly reliant on screens and algorithms. For example, an elderly patient might worry that a doctor is paying more attention to an AI recommendation than to listening to them. They want assurance that AI is an assistant, not a replacement – that it will free clinicians to spend *more* time caring and talking by handling routine tasks, rather than erecting a wall of technology. The idea of a “robot doctor” can be intimidating; most patients still want a human in the loop who has empathy and can interpret the AI through the lens of their individual story.
- **Bias and Fairness:** Patients, especially from historically underserved groups, are concerned that AI could be biased or make unfair decisions^[49]. They may have heard that algorithms sometimes work less well for minorities or women if not properly designed. So a Black patient might be wary of an AI risk score dictating their care if they suspect it might underestimate their needs due to biased training data. Similarly, some could fear discrimination – e.g., could AI deem someone an “expensive” patient and limit their care (this touches on ethical concerns about cost-optimization possibly conflicting with patient interest). They want AI to be a tool for health equity, not something that inadvertently deepens disparities. Transparency about how AI makes decisions is crucial to alleviate this – patients say things like, “I don’t want a computer to decide I’m not worth treating aggressively because of some statistic.” Thus, explaining that AI is checked for bias and is meant to complement, not deny, care is important.

- **Accuracy and Safety of AI Decisions:** There is a general concern about AI making mistakes – “What if the AI is wrong?” Many patients and caregivers are not automatically trusting of technology; they need proof that these tools are accurate and won’t lead them astray. If an AI misses something important or gives a false alarm, it could harm them. For example, if an AI reads an X-ray and misses a fracture, that’s a worry; or if an AI recommends a treatment that turns out suboptimal, who is accountable? Patients want to know that AI systems are rigorously tested and that there’s oversight. Caregivers might ask, “Will the doctor double-check what the AI says?” The concept of **accountability** is key – knowing that the physician or institution still takes responsibility and that there are processes if the AI is wrong (like second opinions or the ability to override).
- **Understanding and Consent:** Patients may feel anxious if AI is used in their care without their knowledge. Some worry about being “experimented on” by unproven tech. They want to be informed that AI is involved and ideally to agree to it. This is both a concern and a desire – they want to be part of the decision to use AI, similar to how they consent for treatments. Also, they often won’t understand how an AI works (and many don’t need the technical detail), but they at least want a basic explanation in plain language of why the AI’s suggestion is being made. Lack of understanding can breed mistrust or even fear of the unknown.
- **Cost and Access Concerns:** Some patients wonder, will AI-driven care cost me more? If a fancy AI recommends some high-tech test or sends me to a big hospital, can I afford it or will insurance cover it? Conversely, they might fear that AI is being used by insurance to cut costs at the expense of care (like an AI might deny them a procedure by deeming it unnecessary). This interplay of cost is on their minds. They want AI to reduce their out-of-pocket costs (maybe by avoiding unneeded procedures), not to be an extra charge or a gatekeeper for coverage. We’ve heard patients appreciate when our trial matching finds them options that could be free via a trial, thus saving them potential huge drug costs – that’s a positive of AI linking to trials. But they would dislike if an AI recommended against a treatment purely on cost grounds.

Addressing these concerns is vital for acceptance. In our practice, we emphasize that patients remain in control – they consent to our services and data use, and we keep them informed. For example, ACS ACTS explicitly states it’s an “AI-driven clinical trial matching service” but under the guidance of ACS cancer information specialists^{[46][50]}, blending AI and human support to maintain trust. We take privacy seriously; patients trust ACS and by extension our system because we uphold confidentiality and they actively sign up for it. Ensuring **trust and transparency** is foundational.

In designing AI in clinical care, we should involve patient input early. HHS could encourage user-centered design, including patients in testing. When patients see AI solving their problems – like reducing wait times, simplifying their journey, or offering hope via new options – they become enthusiastic. But if they perceive AI as an imposed, opaque system that might compromise their privacy or reduce human care, they will resist.

To sum up, patients and caregivers hope AI will *make care more accessible, personalized, and coordinated*, helping them navigate complexity and catch problems early. They want it to amplify the good (better outcomes, more knowledge) and minimize the bad (bureaucracy, errors). On the flip side, they worry about *privacy, losing human connection, bias, safety, and agency*. Any strategy to implement AI at scale should incorporate safeguards and education around these points: robust privacy protections, maintaining the clinician-patient relationship at the core, demonstrable fairness and thorough validation of AI tools, and open communication/consent. If we get this right, patients will not just accept AI – they will *demand* it, because it will be clear it serves their needs and respects their values.

Question 10. Priority Areas for AI Research and Evidence – Literature on AI Impact and Economic Aspects

Q10: “Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?”

a. Are there published findings about the impact of adopted AI tools and their use in clinical care?

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?”

Priority Areas for AI Research (to Accelerate Adoption in Clinical Care):

HHS should prioritize research that addresses the gaps between promising AI technology and its practical, effective implementation in healthcare. Based on the current landscape, we recommend focusing on:

1. **Implementation Science for AI in Healthcare:** It’s not enough to invent AI algorithms; we need to understand how to integrate them into real-world clinical settings successfully. HHS should fund research on best practices for deploying AI in various environments (large academic hospitals, small rural clinics, etc.), including workflow design, change management, and training methods. This research should capture lessons from both successful and failed implementations to create playbooks for health systems. For example, studying how an AI sepsis alert was rolled out across multiple hospitals – what adoption strategies led to usage and improved outcomes versus where it was ignored – will yield general principles. The outcomes could be guidelines or toolkits for health organizations. This directly

accelerates adoption by turning AI from a theoretical tool into one that is effectively used. HHS's RFI itself mentions strengthening **implementation science and best practices, especially for complex scenarios**[\[17\]](#), highlighting this as a recognized need.

2. **Evaluation and Monitoring Frameworks (Safety, Efficacy, Bias):** Research should focus on developing standardized methods to evaluate AI tools (as we discussed in Q4). This includes metrics for clinical impact, user trust, and economic outcomes, as well as processes for continuous monitoring post-deployment. Part of this could be creating open datasets or simulation environments for testing AI (essentially science around how to test AI well). If HHS invests in these, it gives regulators and adopters confidence in AI, speeding up adoption. It also helps sift out ineffective or harmful tools early. An example: researching how to do randomized controlled trials or other study designs for AI interventions (there is ongoing work on cluster RCTs for AI, or adaptive trials where AI is the "intervention").
3. **Bias Mitigation and Fairness in AI:** Prioritize research into algorithms that detect and correct bias, as well as methods to ensure AI tools are equitable. This includes creating diverse training datasets and novel techniques like transfer learning or federated learning to include data from underrepresented groups without breaching privacy. Also, sociotechnical research on how biases manifest in AI-driven workflows and how to mitigate them through both technical and organizational means. The goal is AI that works for *all* patient populations. For example, HHS could sponsor research to improve AI for maternal health in Black women, given disparities, or AI in rural health contexts. Published findings have shown biases (like the care management algorithm example[\[49\]](#)); now research must figure out how to systematically avoid such biases. This is crucial for trust and broad utility of AI, which underpins adoption.
4. **Interoperability and Data Sharing Models:** Research how to safely and effectively share data for AI development – including privacy-preserving analytics, federated learning, and the creation of high-quality synthetic data. HHS should prioritize research that enables multi-center AI studies without violating privacy (e.g., federated learning pilots across healthcare systems in a network). Also, research on the **standardization of health data for AI** (like refining FHIR for AI needs, or extending mCODE) is important. This can inform policy and standards to make data more usable by AI, which as we established will fuel AI advancement.
5. **AI for Public Health and High-Burden Conditions:** HHS might focus on AI research in areas that are national priorities: for example, chronic diseases that drive cost (like diabetes, heart failure), mental health (like AI for suicide prevention or substance abuse support), aging-related conditions (frailty, dementia as mentioned[\[38\]](#)), and health crises (like pandemic response). By prioritizing these, HHS ensures AI is developed where it can have broad impact. For instance, AI research in geriatrics to predict and prevent falls in elderly or manage polypharmacy could greatly improve quality of life and reduce costs.
6. **Human-AI Interaction & Explainability Research:** To enhance trust, research should delve into how AI can effectively communicate with clinicians and patients. This means studying different explanation approaches (visual aids, confidence levels, etc.) and how they affect decision-making and outcomes. Also, research on how clinicians make use of AI recommendations (when do they accept vs override, and why) can guide design improvements. This human factors angle is sometimes overlooked but is vital for safe and accepted AI.
7. **Economic Impact Analysis of AI in Healthcare:** We need rigorous health economics research on AI – how it shifts costs and benefits across stakeholders. This includes micro-level (does adopting AI in X department save money or shift labor?) and macro-level (if AI reduces hospitalizations for heart failure by 20%, what is the net system effect?). Also, "transfers" of costs – e.g., does AI reduce provider costs but increase patient or payer costs, or vice versa? HHS should fund studies and models that simulate these scenarios. This evidence will inform reimbursement and investment decisions. It aligns with question part (b) on costs and benefits.

Now, addressing parts (a) and (b):

(b) Published Findings about the Impact of Adopted AI Tools in Clinical Care:

Yes, there is a growing body of published evidence, though it varies by domain and is still maturing. Some notable findings from literature include:

- **Improved Clinical Outcomes:** Certain AI implementations have shown measurable improvements. For example, a published study on AI sepsis alerts (like one from Duke or Johns Hopkins) found increased early detection and reduced mortality or ICU stay when the AI was used properly[\[34\]](#). In medical imaging, multiple studies (some in *Lancet Digital Health* or *Nature Medicine*) have shown AI algorithms matching or exceeding radiologist performance for detection of cancers (like breast cancer in mammography) or other findings, potentially leading to earlier diagnosis. Another area is in

stroke care: an AI (like Viz.ai) that flags strokes on CT scans has been shown to reduce time-to-treatment (door-to-needle time), which is crucial for outcomes. These findings illustrate impact in terms of speed and accuracy of care.

- **Enhanced Efficiency and Provider Experience:** The literature documents that AI can streamline operations. For instance, a study in *JAMA Network Open* reported that an AI scribe technology led to significant reduction in clinician documentation time and improved work-life balance, albeit without immediate financial ROI[37]. Another example: studies on predictive analytics for no-shows have allowed clinics to double-book strategically and reduce unused appointment slots, effectively increasing throughput without added cost. These are documented in healthcare operations research journals.
- **Patient Engagement and Satisfaction:** Some early reports show that AI-powered patient engagement tools (like chatbot follow-ups for chronic disease management or mental health) can maintain or even improve patient satisfaction while reducing demand on staff. Published trials of chatbot cognitive behavioral therapy, for example, have shown some efficacy in reducing mild anxiety or depression symptoms – not replacing therapists but augmenting scarce resources.
- **Clinical Trial Efficiency:** In research context, publications (including reviews like one in *Journal of the American Medical Informatics Association*) have noted positive outcomes of AI in trial recruitment and design. For example, one review found “AI-powered patient recruitment tools improve enrollment rates by 65%” on average[51], which is a significant impact on how fast trials can complete. Also, predictive models achieving ~85% accuracy in trial outcomes (as reported in summary points of some reviews)[52], hinting that AI could optimize trial protocols.

However, the literature also notes failures or neutral results: e.g., some studies where an AI did not significantly improve primary outcomes (perhaps due to user compliance issues). One article in *NEJM* (Jan 2020 by Wong et al.) concluded that despite hype, there were few AI RCTs and mixed results, suggesting need for better evidence. But since then, number of studies has grown.

(b) Literature on Costs, Benefits, and Transfers of AI in Clinical Care:

The health economics literature is evolving. Key points from what exists:

- Many studies use **modeling and simulation** to estimate potential cost savings of AI. For instance, a Morgan Stanley report projected massive system-wide savings (like that \$300–\$900 billion by 2050 figure) by extrapolating current efficiency gains across the system[36]. PwC’s analysis predicted over \$1 trillion could shift to digital/AI health by 2035[53]. These are forward-looking and make assumptions, but they provide a macro perspective that AI, if widely used, can bend the cost curve by automating labor, reducing errors, and enabling cheaper virtual care.
- **Microsimulation analyses** have been done for specific use cases. For example, an analysis might look at an AI for detecting diabetic retinopathy: it might factor the cost of the AI screening vs the benefit of preventing blindness and find a favorable cost-effectiveness ratio. Similarly, some cost-benefit analyses have been done on AI for stroke triage, showing that faster treatment yields savings by preventing disability (translating to fewer rehab/nursing home costs, etc.). These often appear in specialty journals.
- Some real-world cost evaluations: A study on a hospital’s AI sepsis program might show the program cost X to implement, but saved Y in ICU days and penalties for sepsis outcomes, yielding net savings. If $Y > X$, ROI is positive. Some such data are emerging in conference proceedings or implementation case studies. On transfers: for instance, if AI reduces hospital admissions, hospitals lose revenue (under fee-for-service) while payers save money – that’s a transfer of benefits that literature notes. Conversely, under value-based care, the hospital as ACO saves costs and benefits. So, context matters.
- **Productivity and labor transfer:** Another angle literature discusses is who benefits from efficiency. If AI cuts down documentation time (benefit to physician time, which could see more patients or go home earlier), the “financial” benefit might not directly appear in a ledger, but it could reduce burnout and staff turnover (which is a cost if they leave). These indirect benefits are discussed qualitatively and increasingly quantitatively (some studies tie decreased burnout to retention rates, etc.).
- The literature calls for more **robust evidence generation**. Recognizing this, journals like *NEJM Catalyst*, *JAMIA*, *BMJ Quality & Safety* have called for publishing not just accuracy of AI but *impact on workflow and cost*. There are now frameworks for assessing health AI similar to drug HTA (Health Technology Assessment), incorporating cost-effectiveness.

In terms of transfers, one key observation in literature: AI might incur upfront costs (software, training, maybe increased imaging if AI finds more nodules that need follow-up) and savings come later (fewer late-stage cancers to treat, etc.). So, the timeline of costs/benefits differ among stakeholders. The literature often frames this in terms of **who invests vs who gains**. For example,

primary care might invest in AI to prevent hospitalizations, but the hospital (if paid per service) loses revenue; however, society/payers gain net benefit. Under capitated models, it aligns. So, research often models scenarios under different payment models.

Given the still emerging state of evidence, HHS should promote more pilot studies and dissemination of results. It might even require reporting of outcomes from AI demonstration projects funded by HHS, to build a public knowledge base.

HHS's Role in Research and Evidence:

HHS should fill the evidence gaps by funding and possibly coordinating multi-site evaluations. For instance, to understand economic impact, HHS could run or fund a demonstration where a certain AI is implemented in several diverse sites with analysis of clinical and financial outcomes.

We note that to date, many promising results appear in single-site studies or retrospective analyses. The next step is more rigorous prospective multi-center studies. HHS is well positioned to drive this via agencies like AHRQ (for safety and quality studies), NIH (for clinical effectiveness trials of AI), and CMS (through demonstration projects or model tests focusing on AI).

Also, HHS should prioritize areas that combine high unmet need, potential for AI to help, and alignment with national initiatives (e.g., Cancer Moonshot, Opioid crisis, rural health). For example, research on AI-assisted tele-oncology for rural cancer patients hits those notes.

Conclusion of Question 10:

In conclusion, HHS should focus research on bridging the “last mile” of AI adoption: implementation science, evaluation methods, bias mitigation, interoperability, and strategic high-need clinical domains. Published evidence so far is promising but not yet comprehensive; it shows AI can improve outcomes and efficiency in specific instances[34][36], but we need broader validation and economic analysis. Literature on cost-benefit suggests significant long-term savings and better care, but also points out that benefits may accrue to different parties than the costs – implying a need for HHS to align incentives (through payment models or policy) so that those investing in AI also share in the gains. By prioritizing research that produces robust evidence and frameworks, HHS will build the foundation of trust and knowledge necessary for widespread AI adoption in clinical care.

Conclusion and Alignment with HHS's Genesis Project Vision

In closing, Massive Bio strongly supports HHS's bold initiative to accelerate AI adoption in clinical care. We have addressed each of the RFI's questions with actionable insights, drawn from our on-the-ground experience as an AI implementation partner and supported by emerging evidence in the field. Our comment underscores several cross-cutting themes:

- **Bold Vision, Pragmatic Action:** We echo President Trump's visionary AI agenda and HHS's transformative goals – AI in healthcare can “*propel the US health care system forward*”[3] – and we pair that vision with concrete steps around regulation, reimbursement, and research. By modernizing regulations, aligning payment incentives, and investing in implementation and evidence, HHS can convert AI's promise into everyday reality in clinics and hospitals across America.
- **Patient-Centric Innovation:** We place patients at the center. AI should not be innovation for its own sake, but a tool to enhance patient and caregiver experiences, improve outcomes, and reduce costs – in short, to **Make America Healthy Again** through smarter care[1]. Whether it's a cancer patient finding a lifesaving trial via Massive Bio's AI, or a rural senior avoiding hospitalization thanks to predictive analytics, these are the human stories AI should create at scale. Our recommendations around interoperability, equity, and trust all serve to ensure AI works *with* patients, not on them.
- **Massive Bio as a National Partner:** We have highlighted how Massive Bio's work – in partnership with ACS, recognition by Cancer Moonshot, and technology like Reticulum Nexus – exemplifies the operationalization of AI at national scale. We break down silos by connecting community clinics, patients, and research sponsors in one intelligent network[44]. We generate real-world evidence and improve trial access for tens of thousands of patients[11][12]. This aligns perfectly with the spirit of the **HHS Genesis Project**, which calls for a historic effort to harness AI for national challenges[31]. Just as DOE's Genesis Mission is integrating data and supercomputing to accelerate science[54][55], HHS's Genesis Project in health should integrate clinical data, AI tools, and delivery systems to accelerate better health outcomes. Massive Bio stands ready as an implementation arm for HHS in the cancer domain (and beyond), bringing our multi-stakeholder ecosystem and proven AI platform to help realize this ambitious mission.
- **Regulation, Reimbursement, R&D Levers – a Coordinated Push:** Our answers stressed that these three levers must work in concert. For example, regulatory clarity (what is allowed and safe) combined with reimbursement incentives (making it financially sensible) and R&D support (building the evidence and methods) will together create an environment where AI flourishes responsibly. We provided detailed suggestions in each area – from updating FDA and HIPAA



frameworks[19][17], to introducing new CMS payment models and quality incentives[20], to funding research in implementation and bias mitigation[17]. If HHS pulls all these levers, the effect will be multiplicative, not just additive.

- **Interoperability and Standards as Enablers:** A recurring enabler in our comment is interoperability. Data must flow safely and efficiently for AI to reach its potential[18]. We advocated for HHS to double down on standards like FHIR/mCODE[10], open datasets, and exchange frameworks. These are the rails upon which the AI engine runs; HHS's ongoing work through ONC and associated initiatives is critical and should be accelerated.
- **Evidence and Trust:** We emphasize creating a virtuous cycle of evidence → trust → adoption. We urged HHS to prioritize research that produces rigorous evidence of AI's impact (clinical and economic)[51][34] and to develop certification and evaluation processes that assure providers and patients of AI tools' safety and efficacy. When frontline clinicians see proof that an AI has been validated and improves care, and when patients see that it's been thoughtfully implemented with safeguards, they will trust and adopt these tools. HHS can facilitate this by being both a catalyst for generating evidence (through grants, demos) and a communicator of that evidence (through guidelines, learning collaboratives).
- **Spirit of Public-Private Partnership:** Achieving national-scale AI transformation in healthcare is a team effort. HHS is wisely seeking broad input – and must continue to engage industry, providers, patients, and academia in crafting solutions. Our comment highlights how Massive Bio, a private sector innovator, can partner with public initiatives (like CancerX, Moonshot, ACS) to deliver results. We believe HHS should continue fostering such partnerships, whether through formal cooperative agreements, innovation challenges, or consortiums. By uniting the entrepreneurial energy of companies with the convening power and oversight of government, we can move faster and smarter.

Massive Bio thanks HHS for the opportunity to contribute to this important dialogue. We are excited by the Administration's commitment to making the United States the world leader in healthcare AI – a goal that is within reach if we act boldly and collaboratively. We are particularly enthusiastic about the focus on high-cost, high-stakes areas like cancer, where AI and data can drastically change outcomes and costs, as we have begun to demonstrate.

In the spirit of implementation and **Genesis**: we urge HHS not only to plan and strategize, but to launch demonstration projects and scale-up efforts – essentially, to **ignite the AI engines** in real healthcare settings across the country. Massive Bio is prepared to be on the front lines of this implementation wave. Whether it's expanding our AI-powered trial access network in partnership with federal programs, contributing data and insights to national research efforts, or helping to develop standards and training curricula, we are eager to help operationalize the RFI's outcomes.

By focusing on high-impact use cases (such as those Massive Bio has encountered in trial matching, patient journey analytics, and multi-modal decision support), and by adhering to the best practices outlined above, HHS can create a benchmarking ecosystem that accelerates innovation while maintaining **rigorous standards of transparency, equity, and reproducibility**. Such benchmarks will build trust in AI solutions – among clinicians, patients, and regulators – and ultimately help ensure that **artificial intelligence delivers reliable, safe, and effective tools** in the fight against cancer. We appreciate the opportunity to contribute our perspectives and are eager to collaborate in the community-wide effort that this RFI will catalyze.

With the recommendations outlined – clear policies, aligned incentives, robust research, and an unwavering focus on patient centered outcomes – HHS can lead a transformation as profound as any in modern medical history. Together, we can ensure that **AI becomes a trusted, ubiquitous part of clinical care** – improving quality, equity, and value in the health system. This is a moment of opportunity; by seizing it, HHS will not only save lives and dollars but also fulfill the promise of American innovation and compassion in healthcare.

Massive Bio is honored to lend our voice and stands ready to collaborate on next steps. Let us move forward decisively into this AI-enabled future of health, turning the visionary Genesis Project into a concrete **Genesis of better care for all Americans**.

We thank you for considering our input and for your commitment to harnessing AI for the public good.

Sincerely,

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